Primary Visual Pathways (I)

Introduction to Computational and Biological Vision

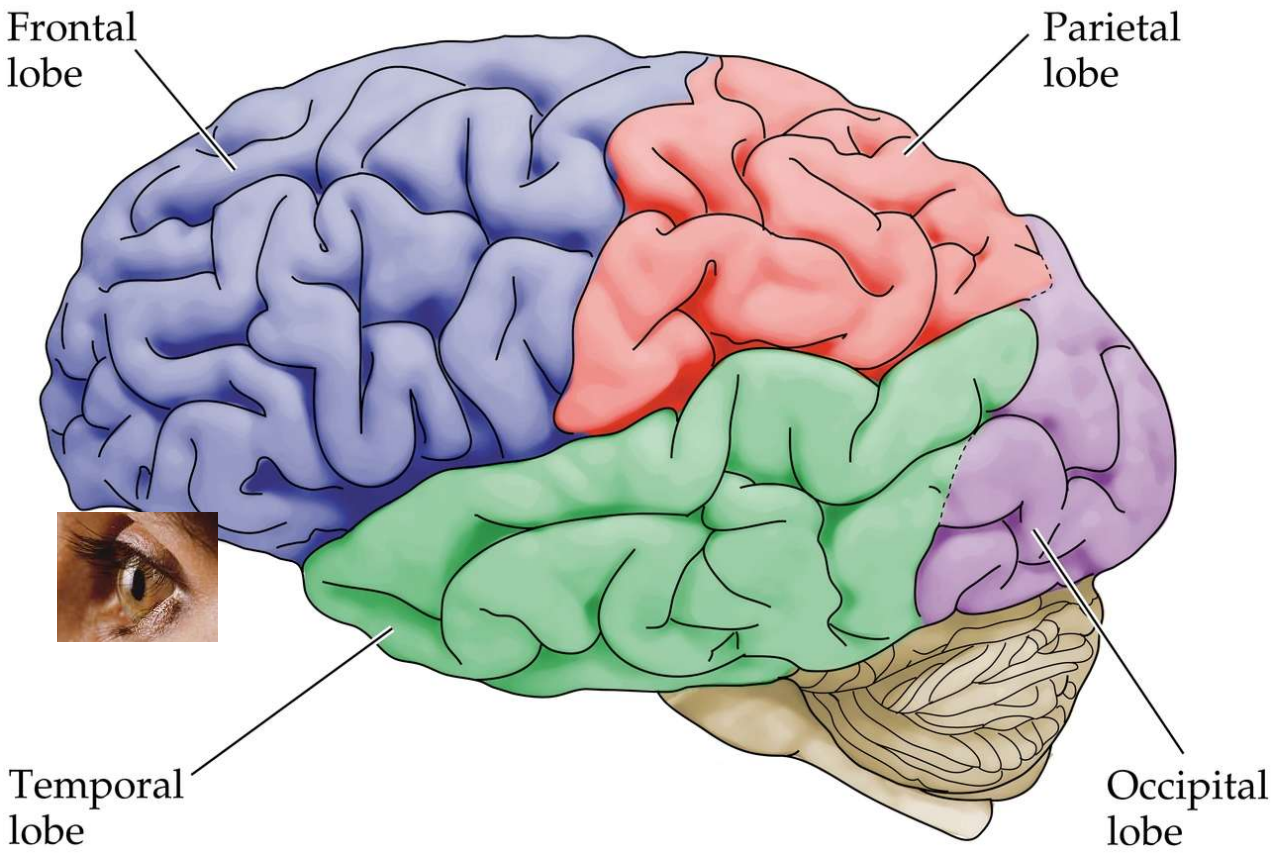
CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

Primary visual pathway

Where does visual information go from the eye?



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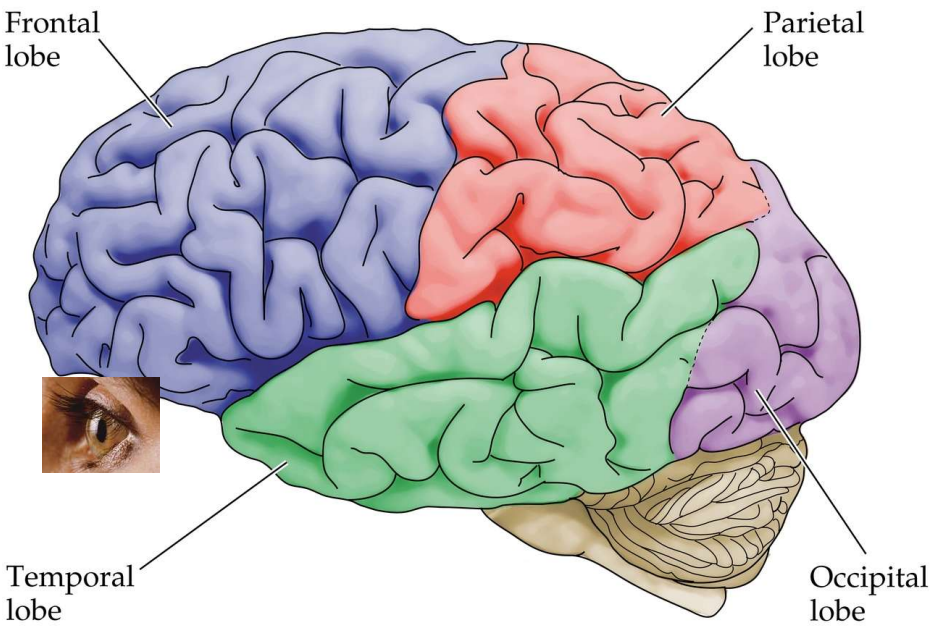

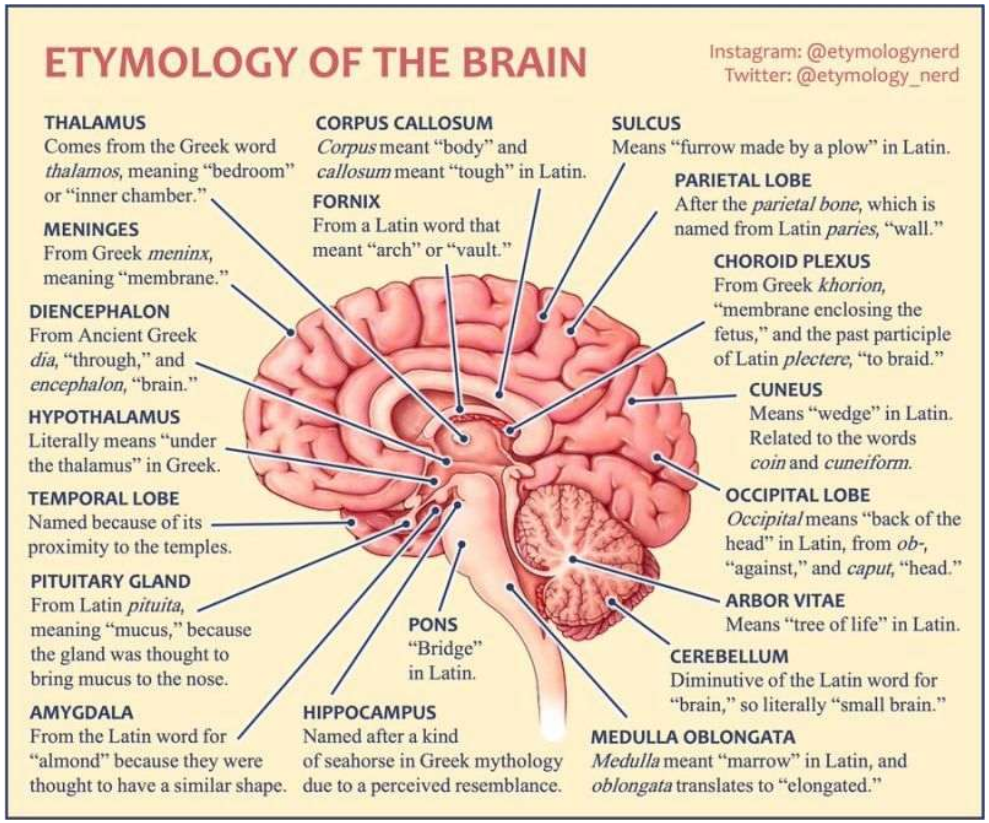

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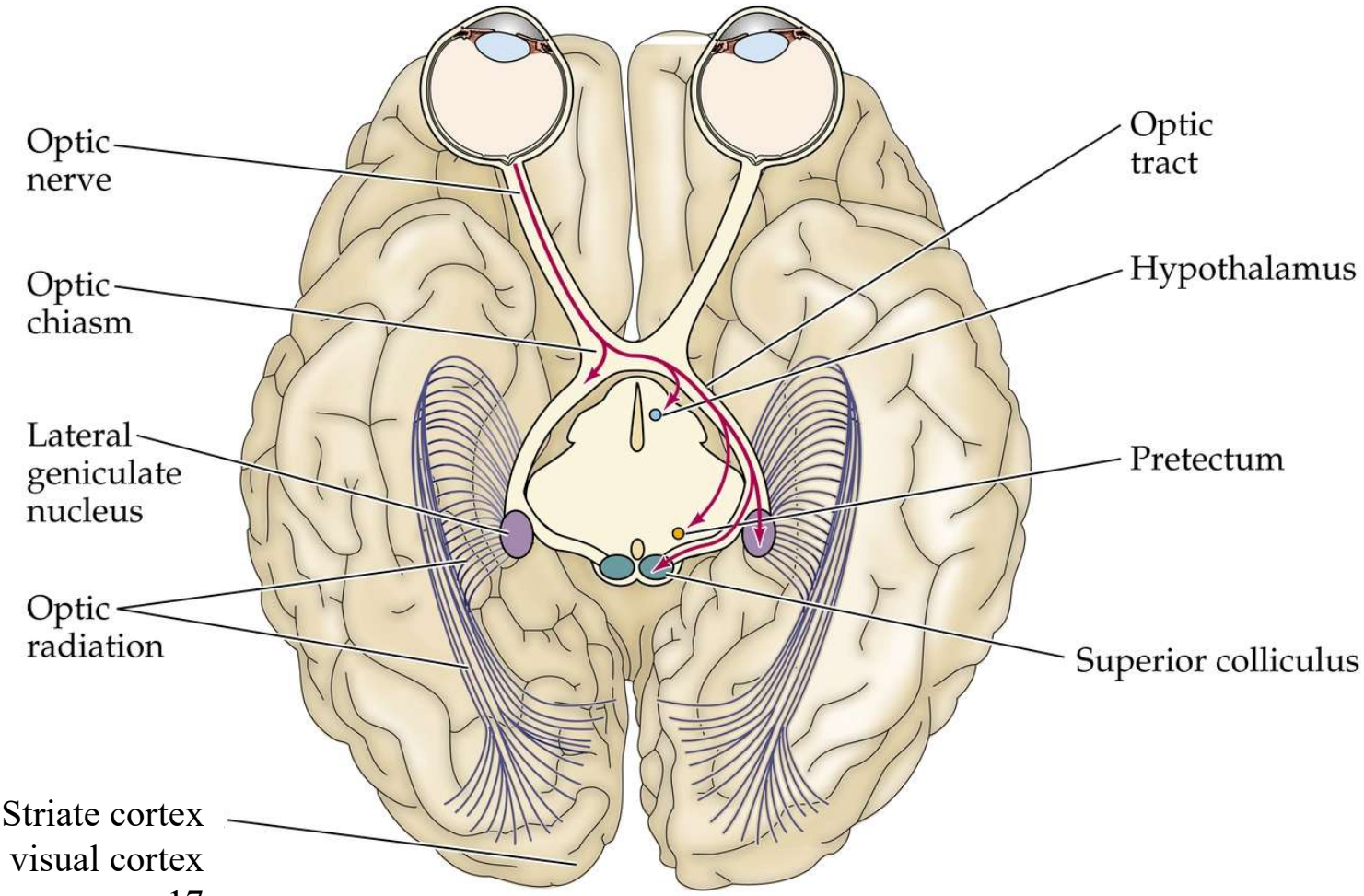

Striate cortex
Primary visual cortex
Brodmann area 17
V1

Primary visual pathway

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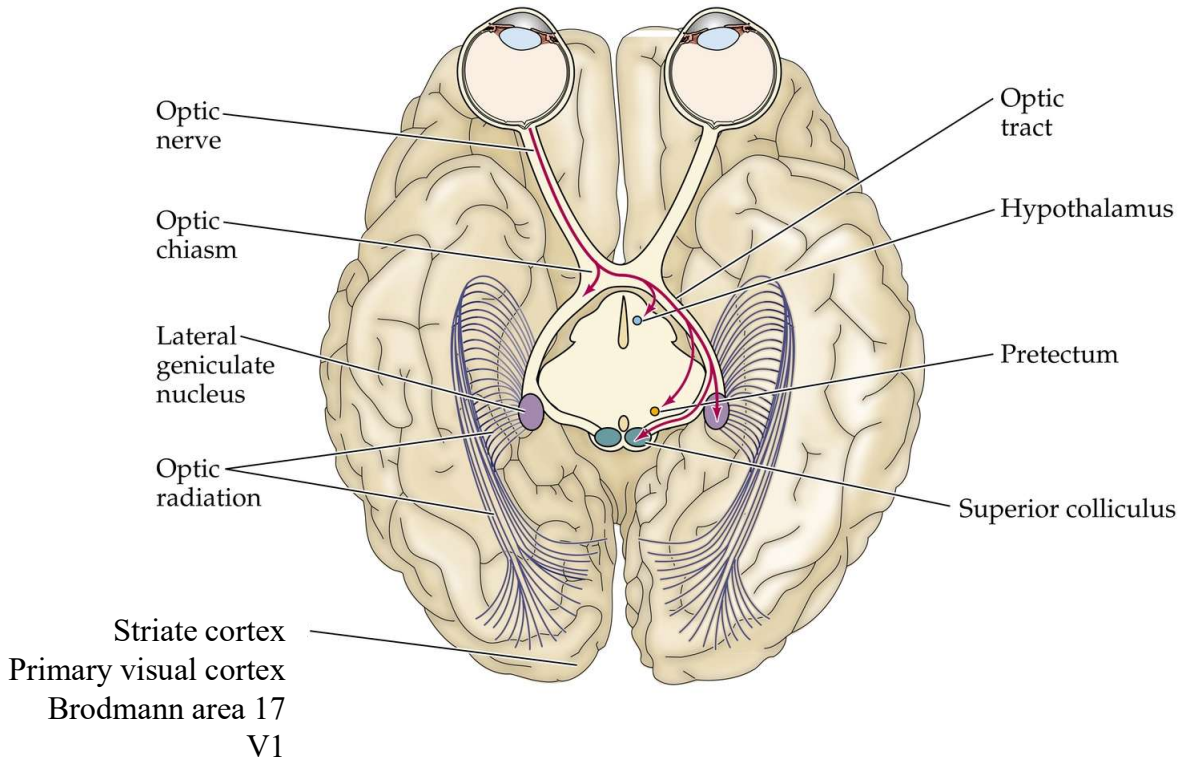
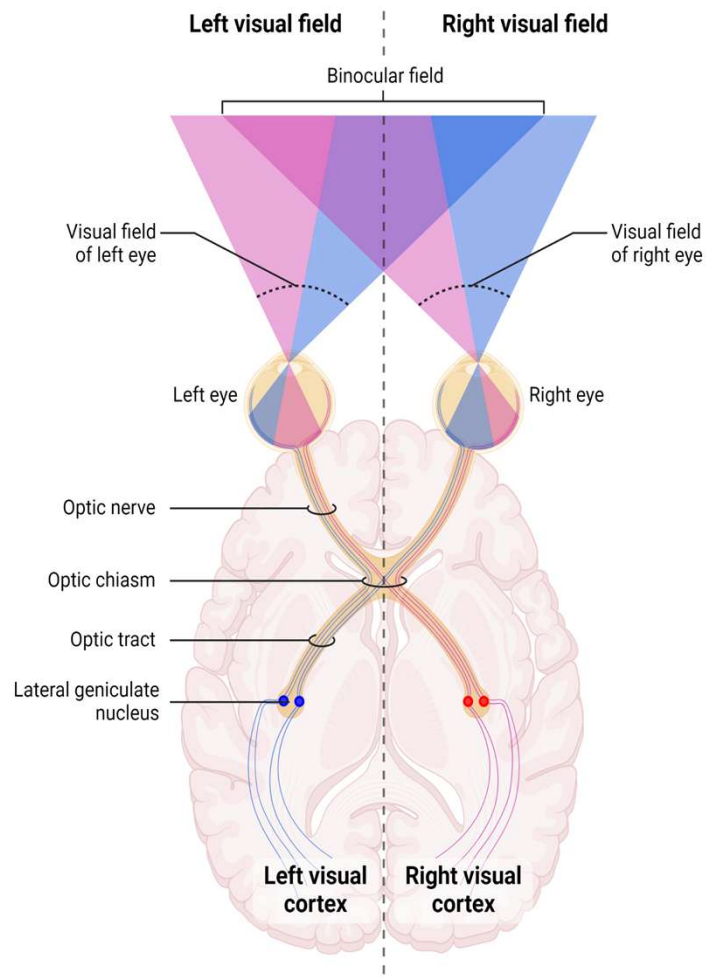
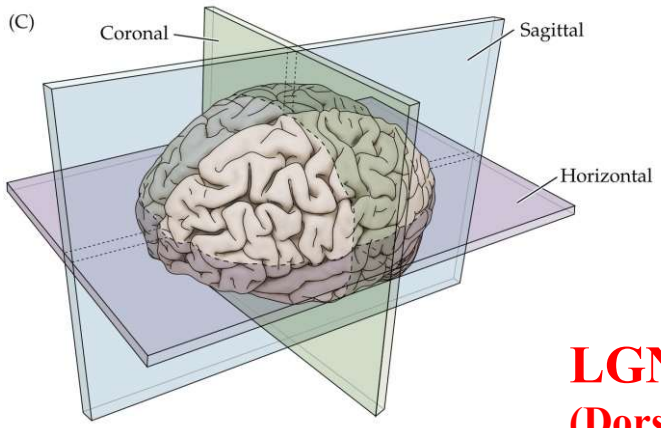




[Human Neuroanatomy, Hutchins, 2024]



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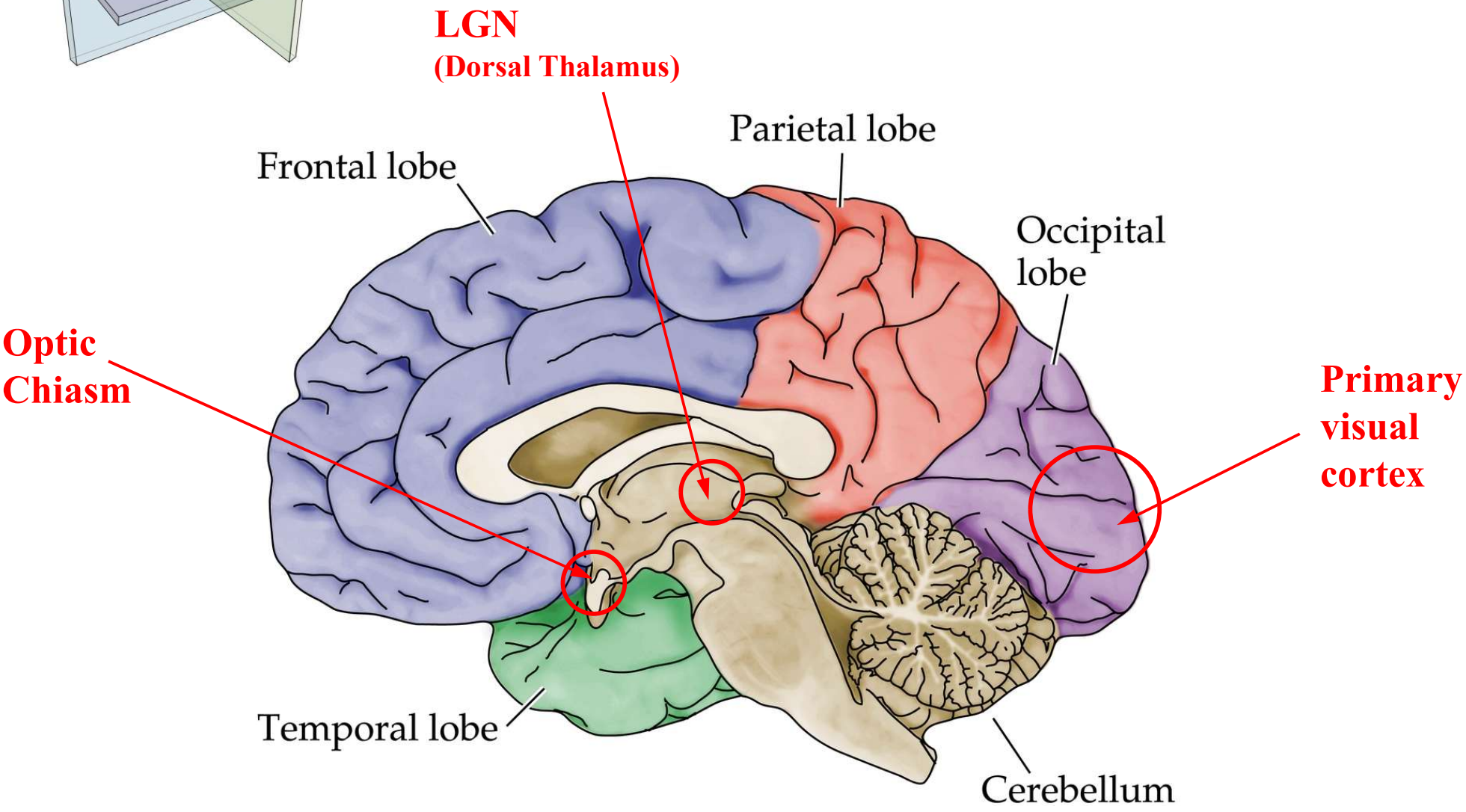

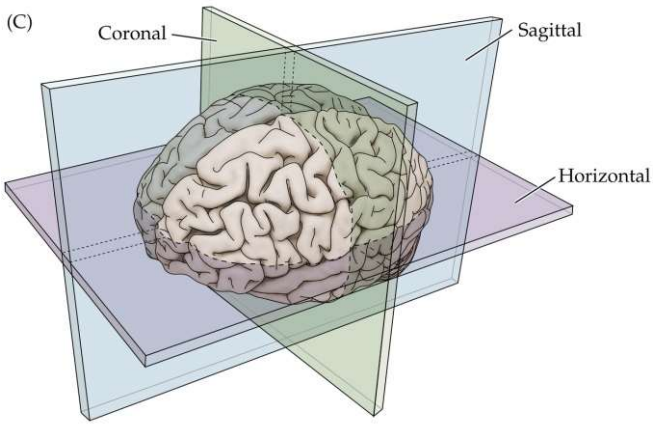

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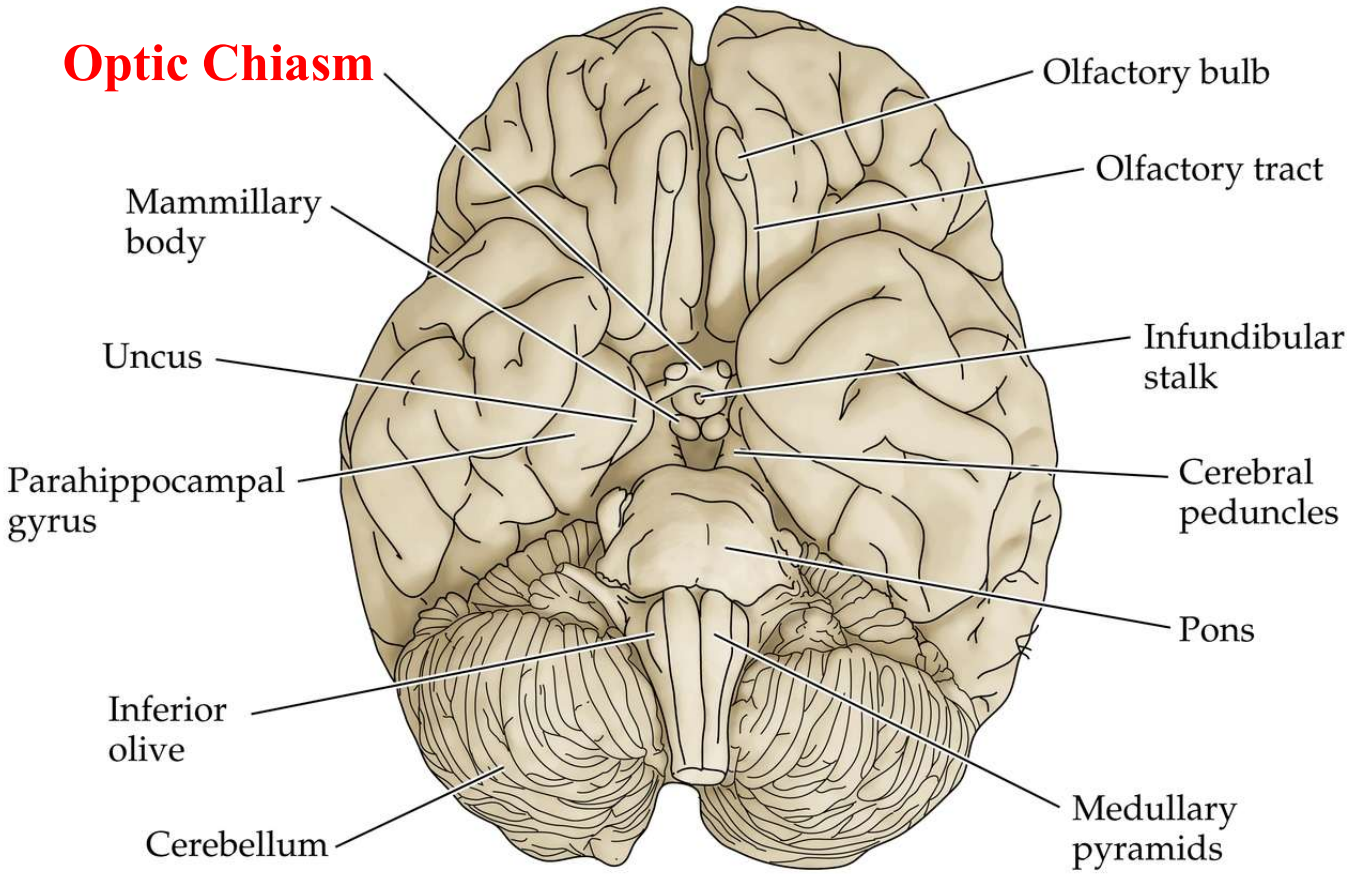

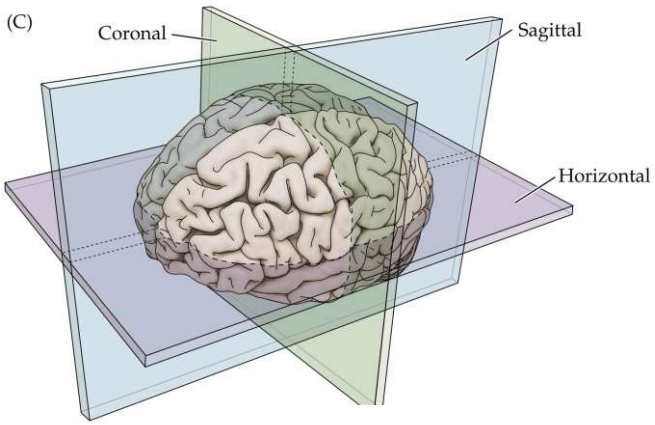

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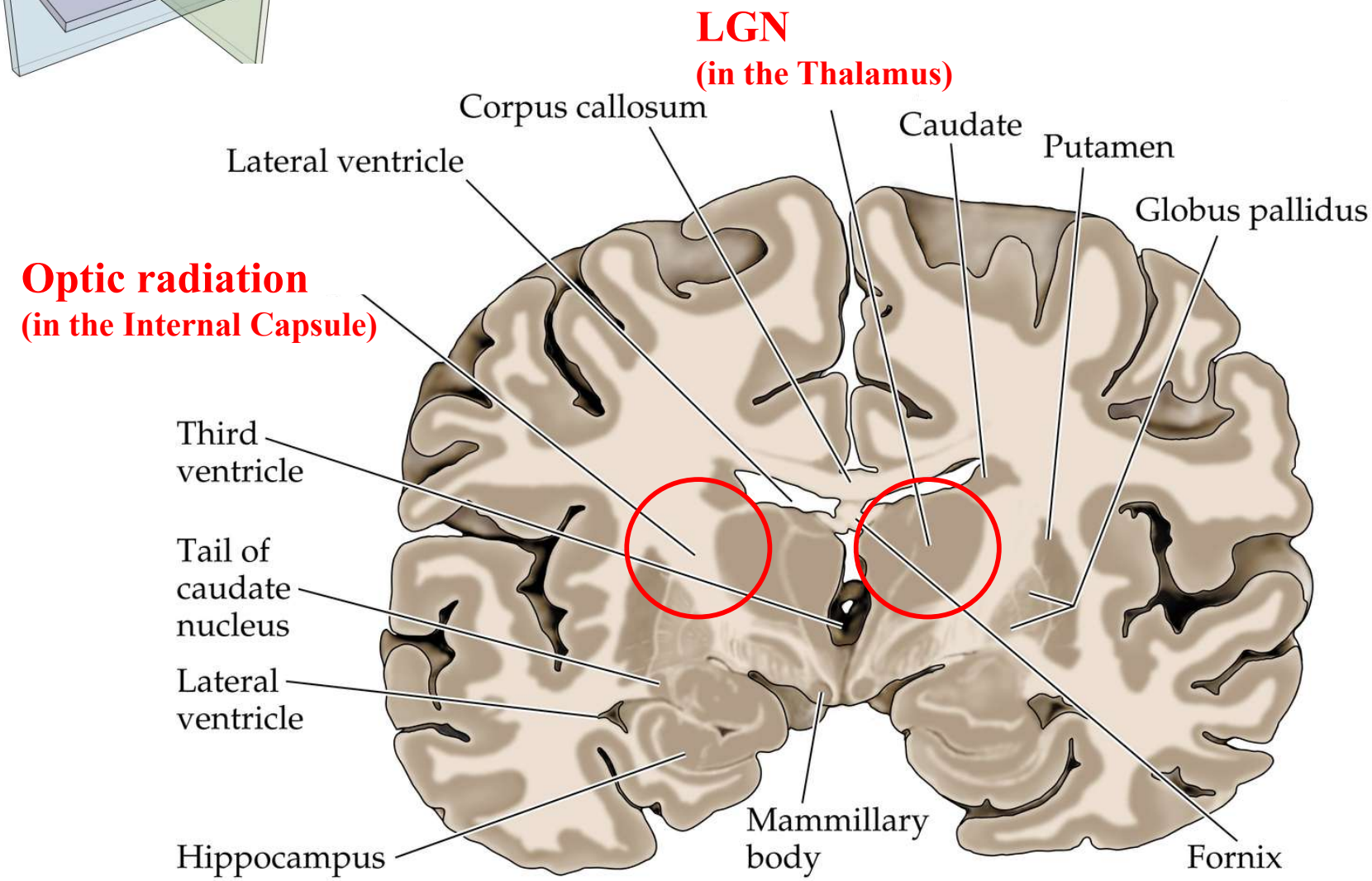

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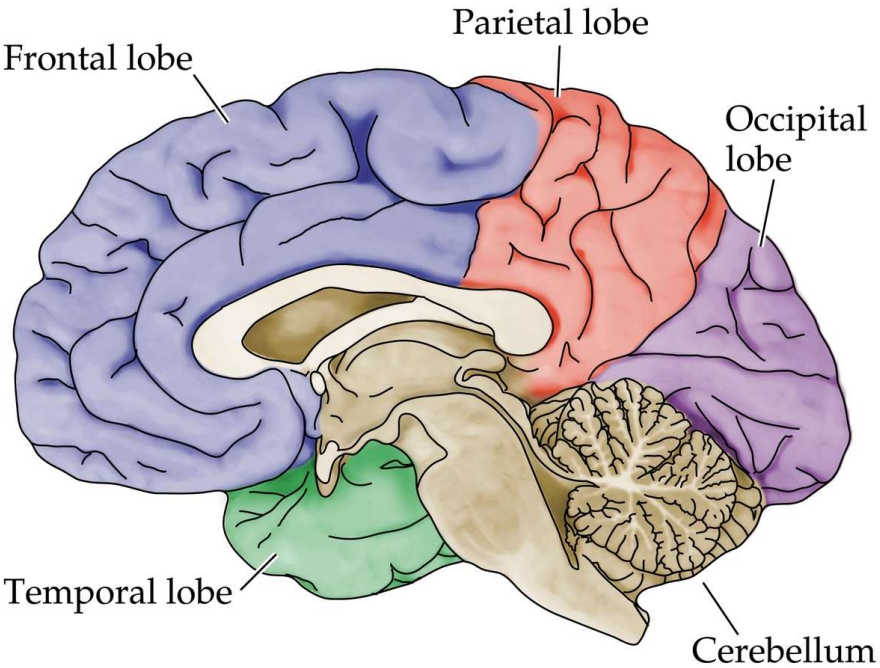
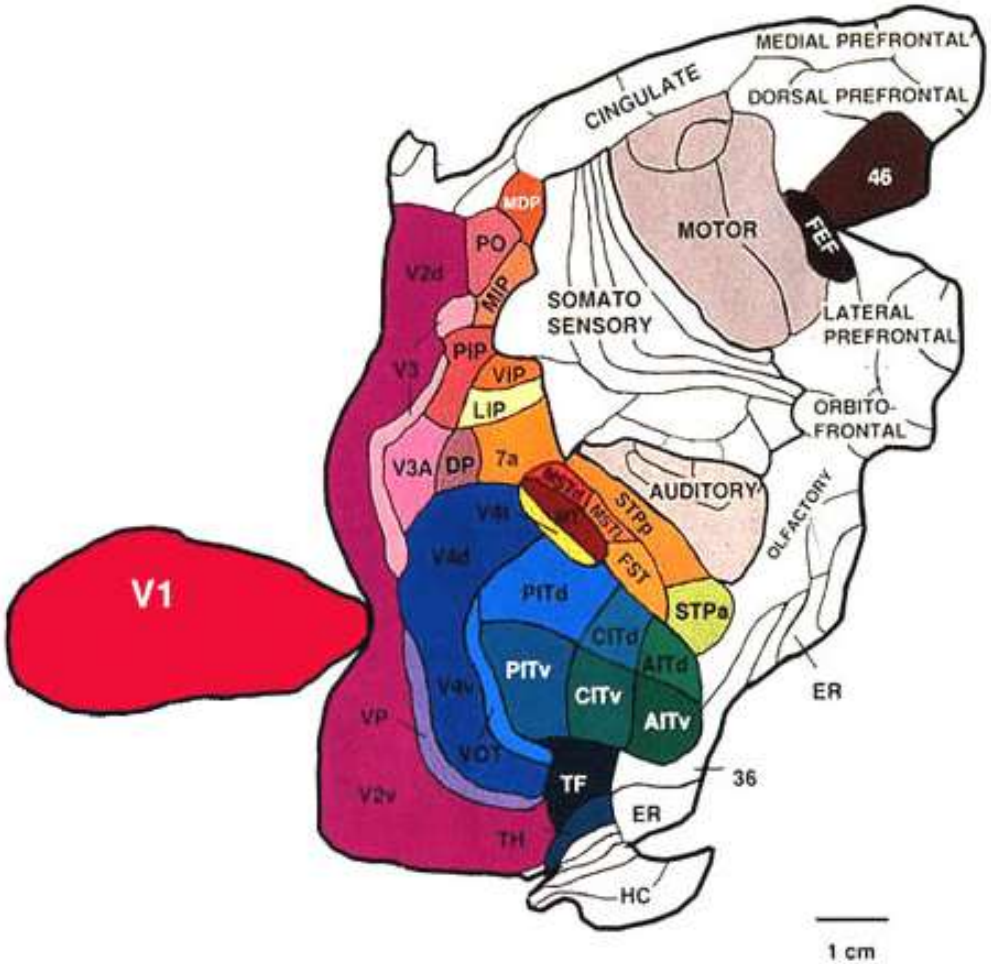
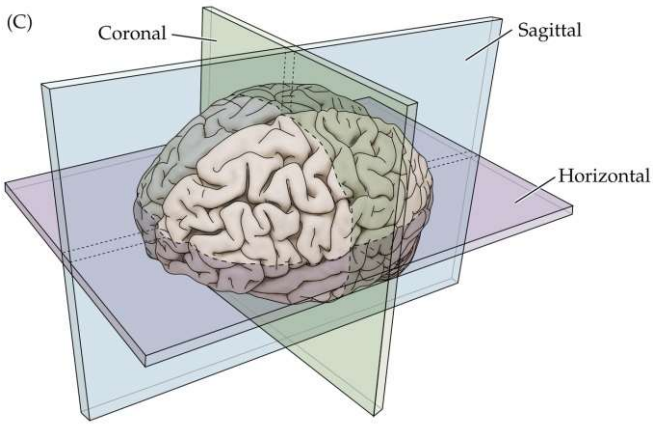




[Van Essen et al. 1992]



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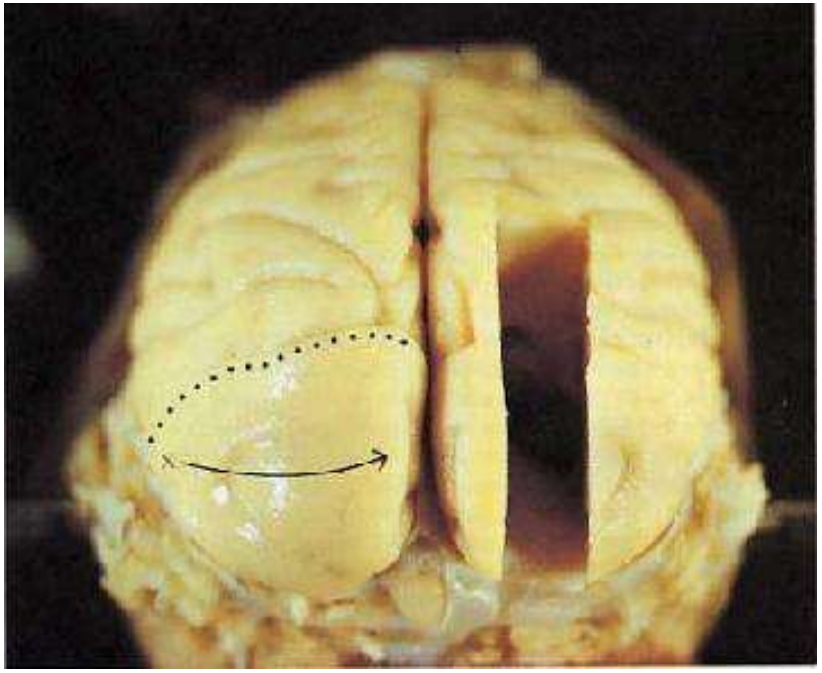

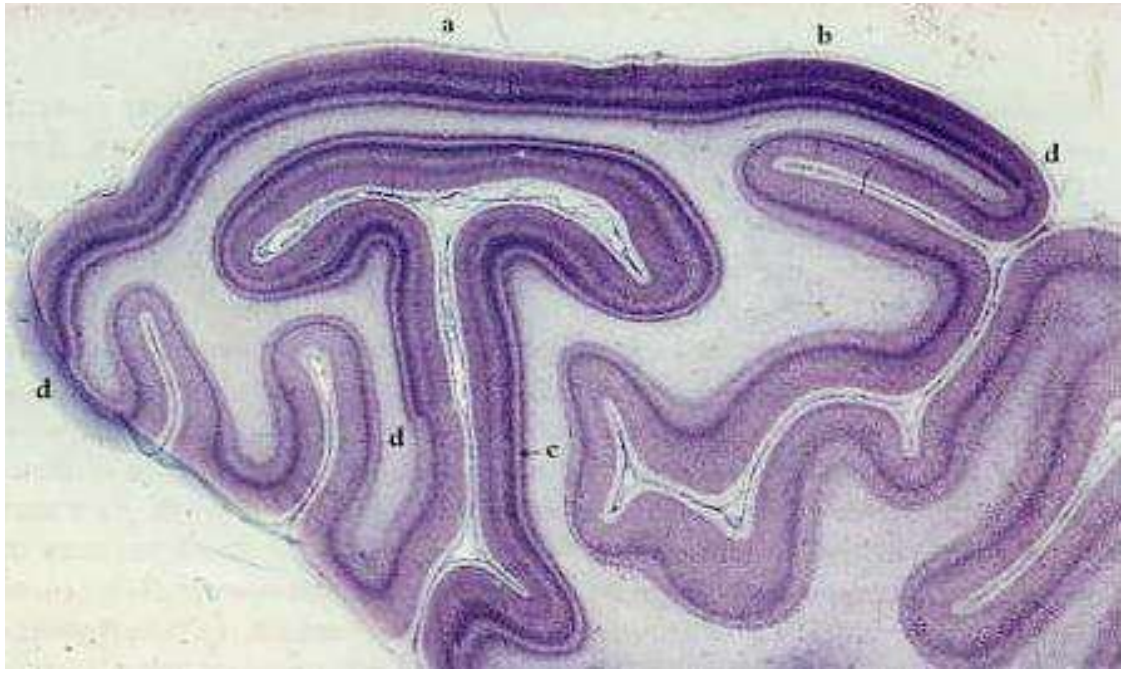

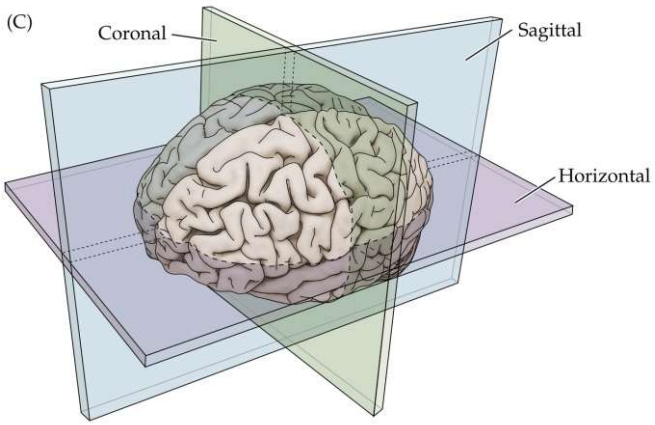

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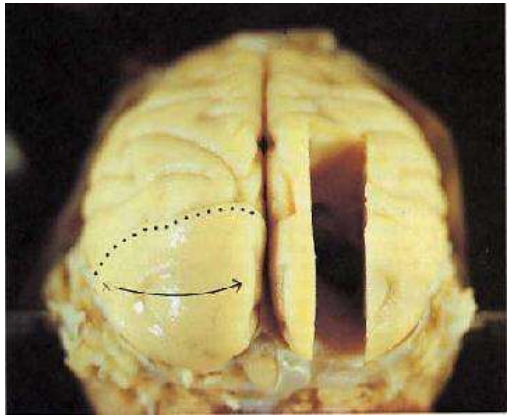

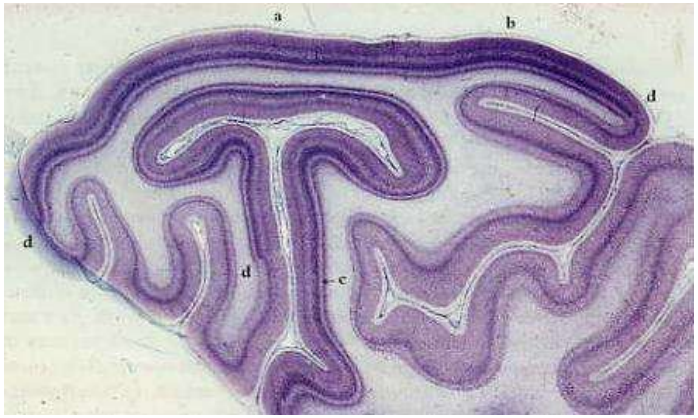

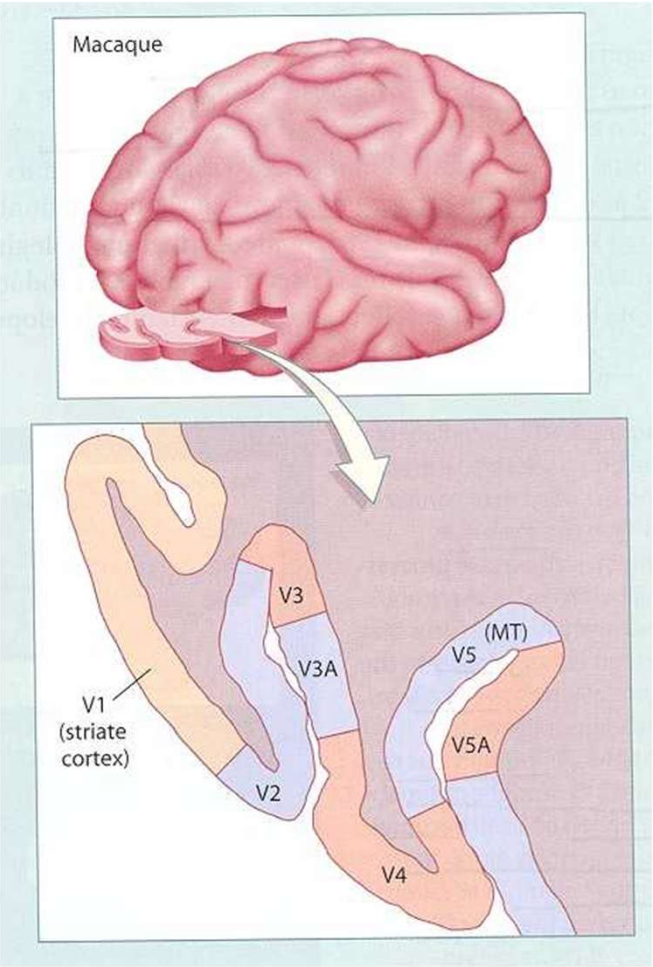

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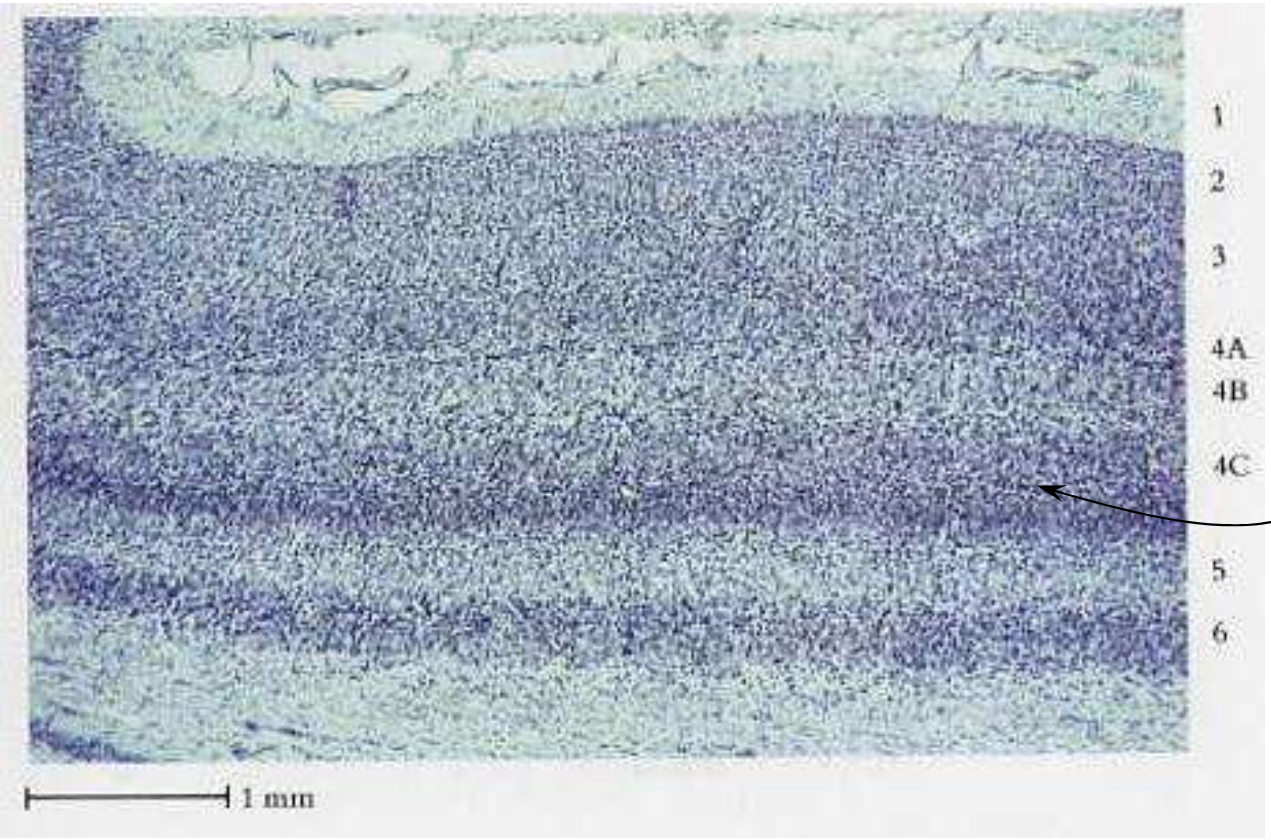


Input from
optic radiation

Primary visual pathway

Be the scientist…

Given this gross anatomy how shall you study the function of this system?

(Hint - think of Hartline…)

Primary visual pathway

LGN cells: Receptive fields and tuning properties

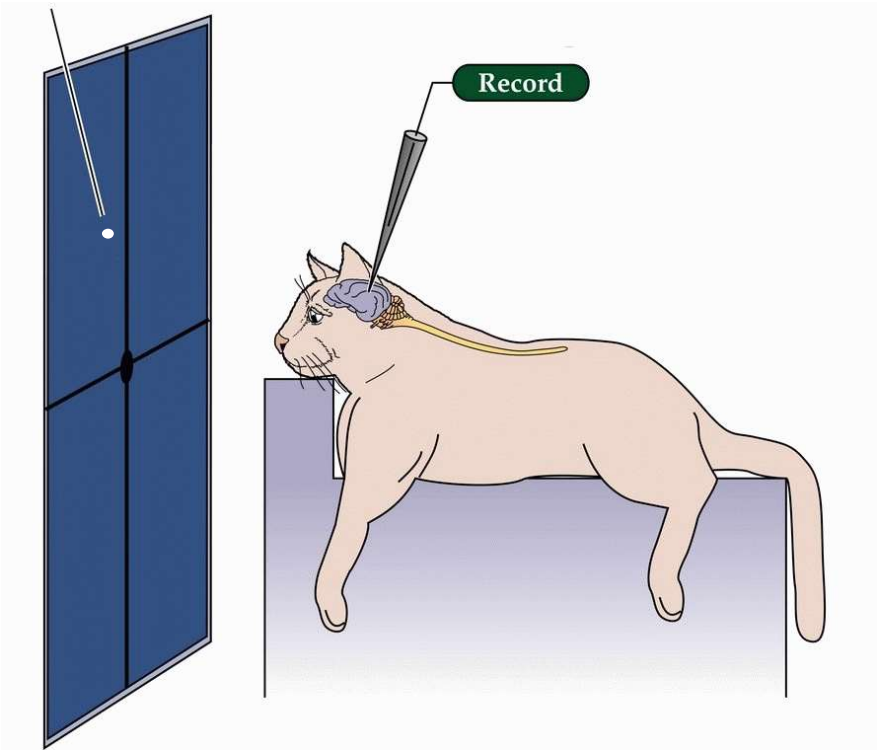
(Classical) Receptive field: A region of the retina (visual field) that must be stimulated directly in order to obtain a response in that cell

[Sherrington 1906]

[Hartline 1938]

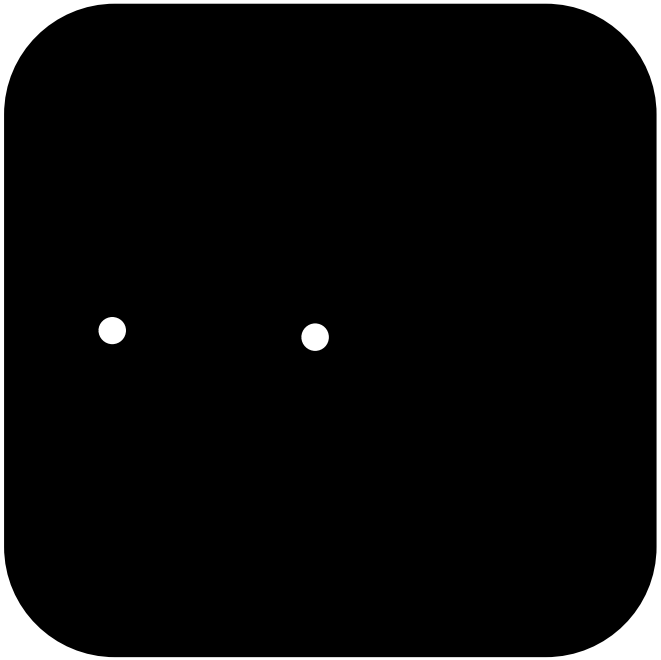
[Kuffler 1953]

Point stimulation



Primary visual pathway

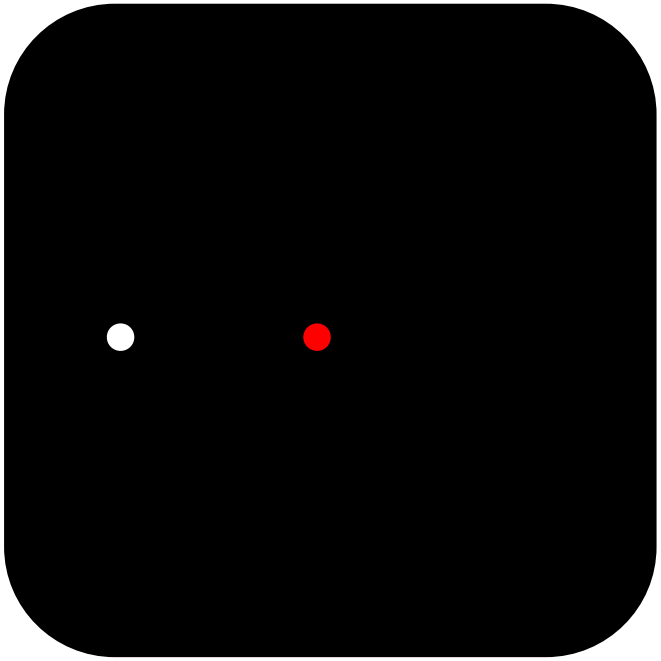
LGN cells: Receptive fields and tuning properties



Visual field (= retina)

Primary visual pathway

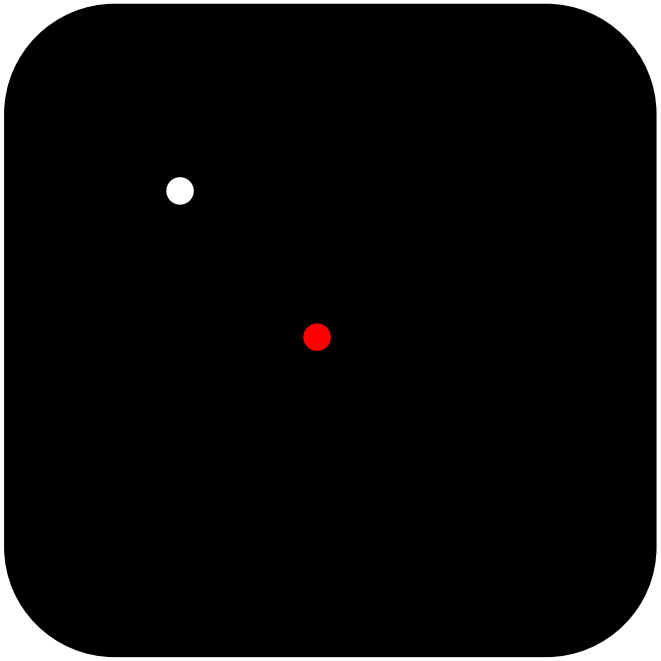
LGN cells: Receptive fields and tuning properties



Visual field (= retina)

Primary visual pathway

LGN cells: Receptive fields and tuning properties

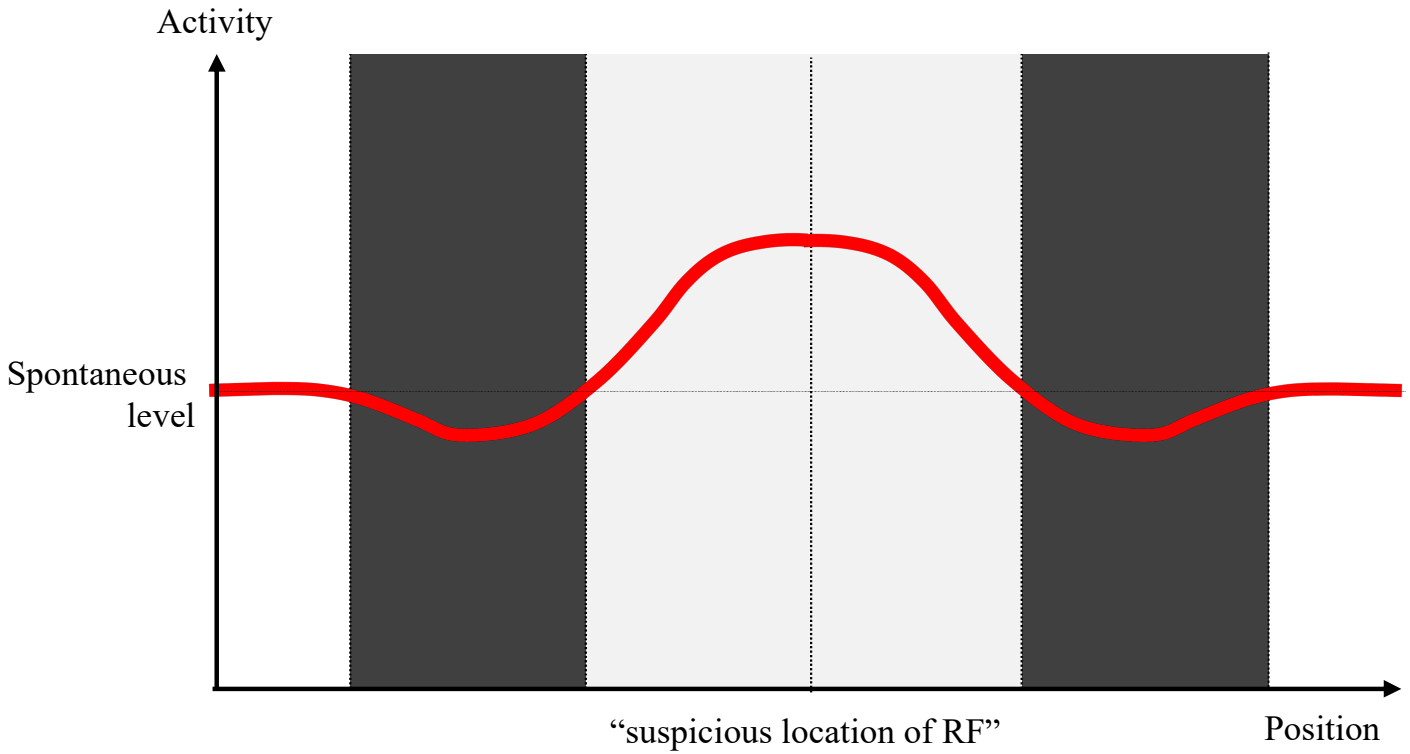


Visual field (= retina)

Primary visual pathway

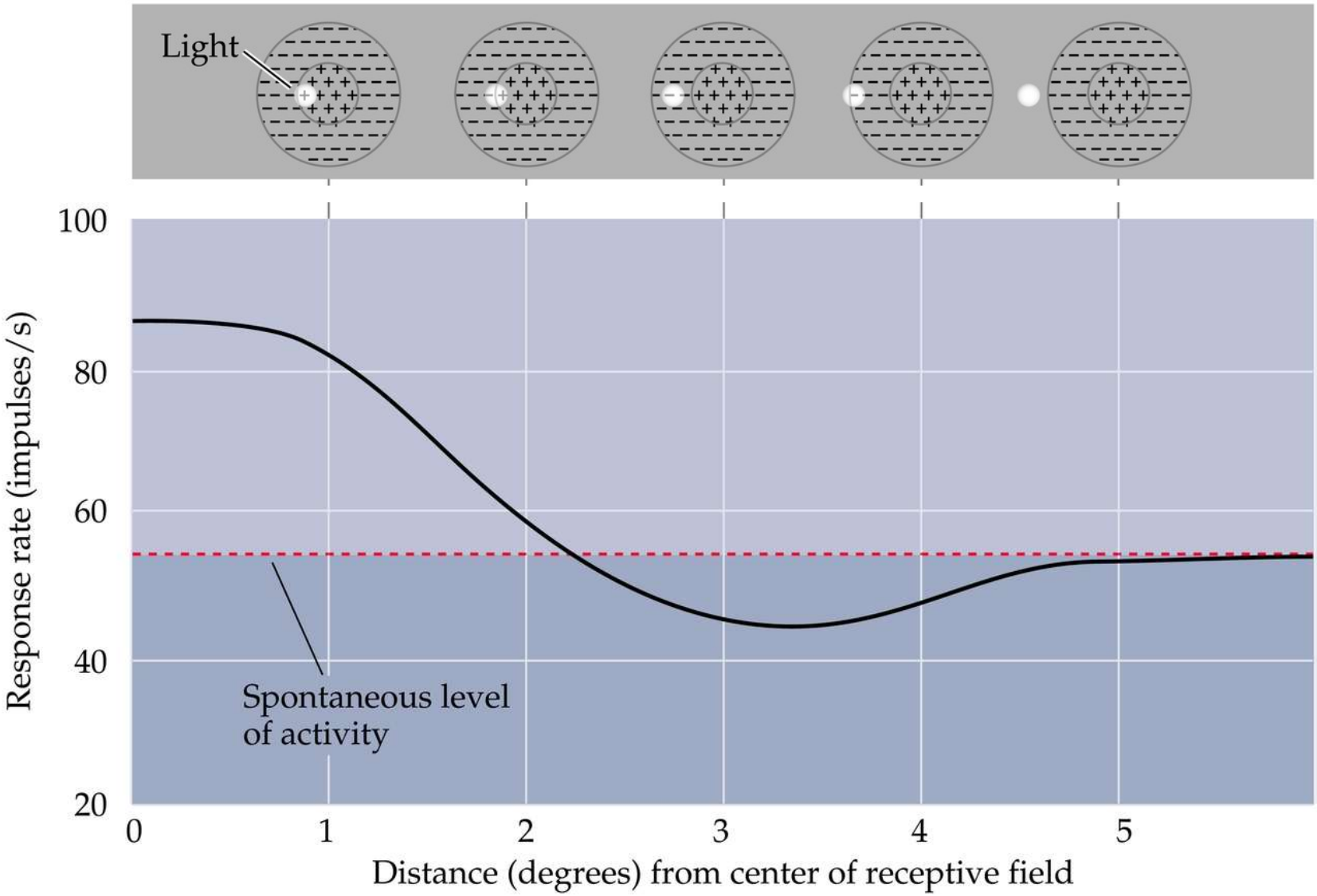
LGN cells: Receptive fields and tuning properties

Visual field (= retina)



Primary visual pathway

LGN cells: Receptive fields and tuning properties



Primary visual pathway

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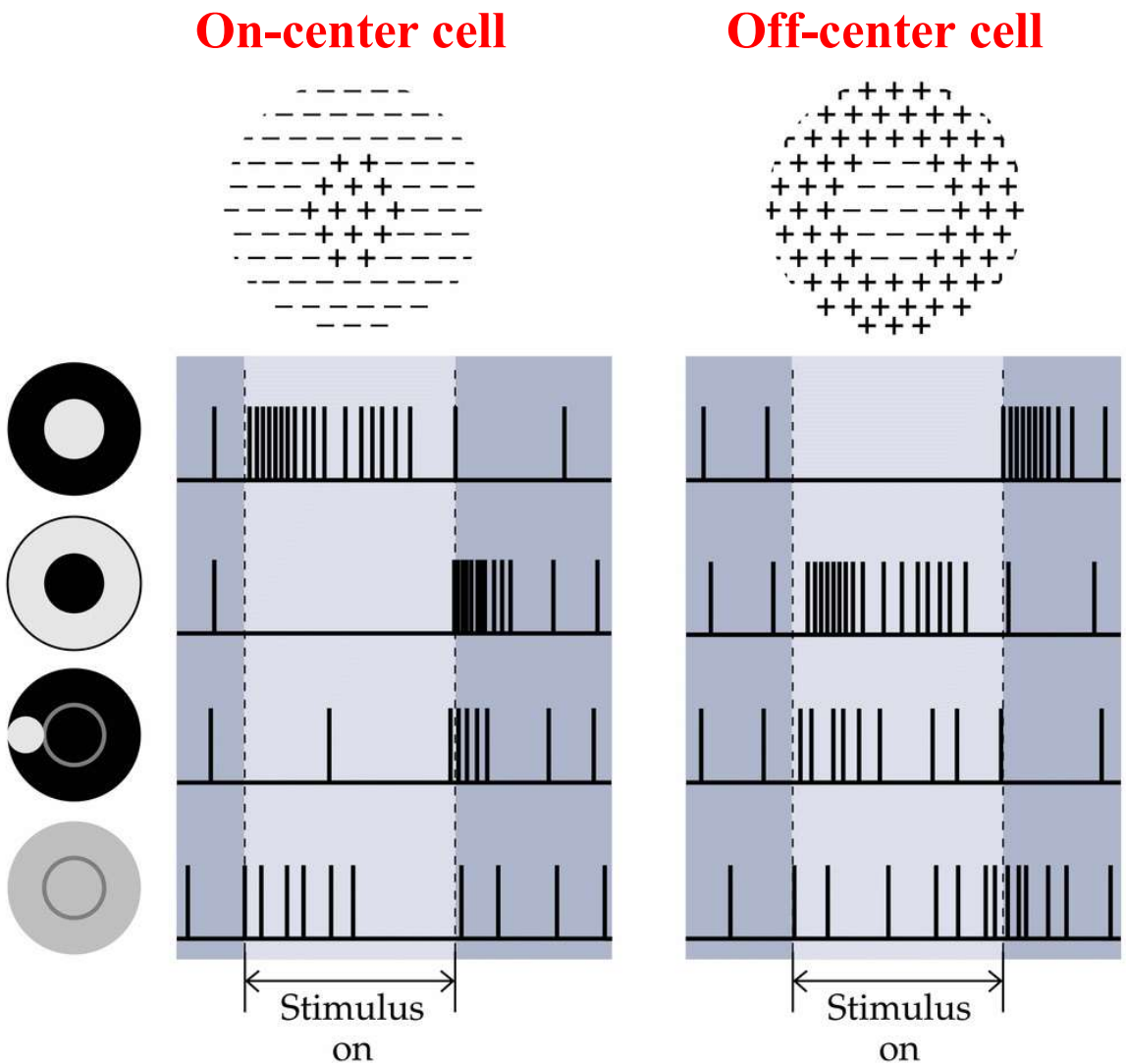

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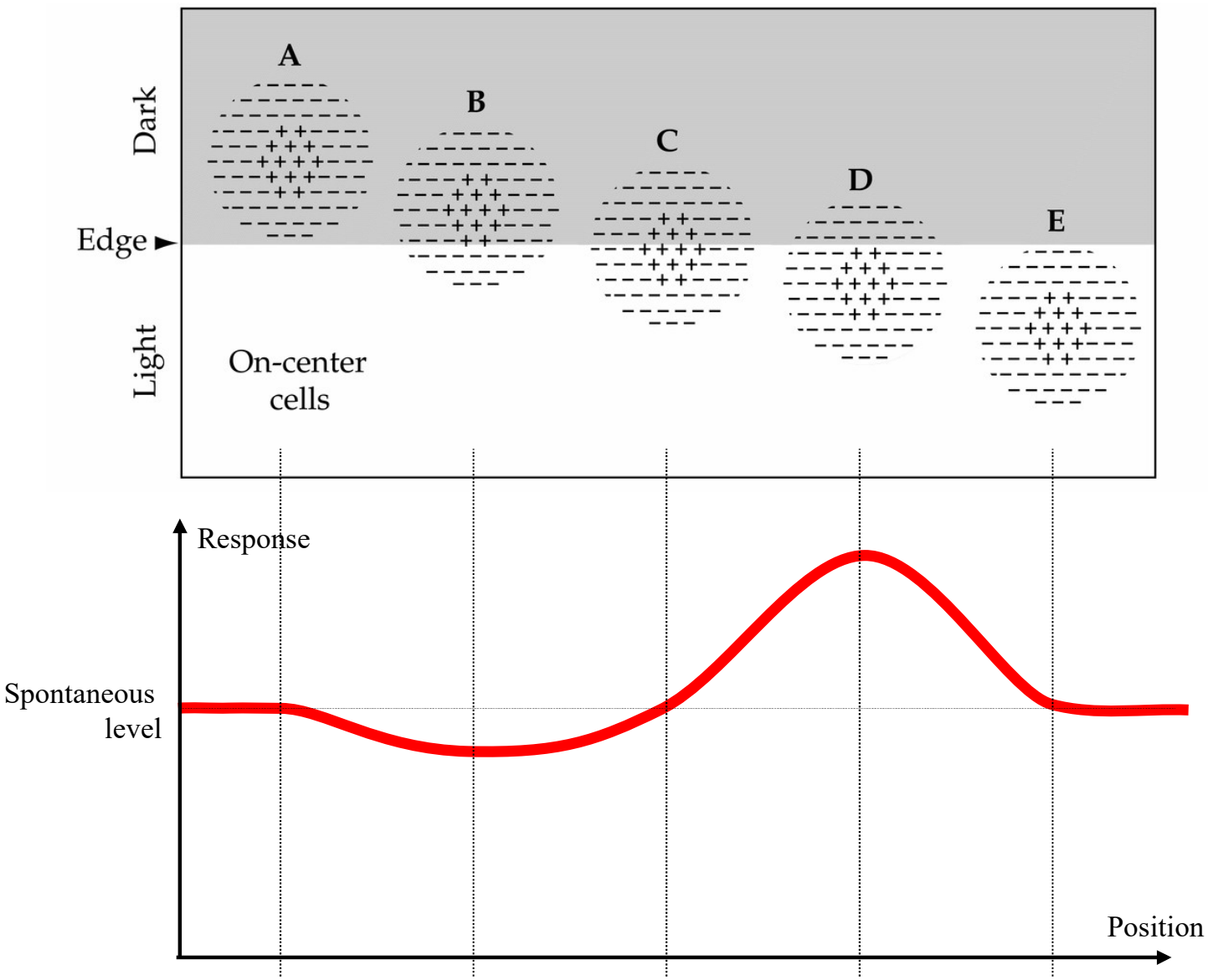
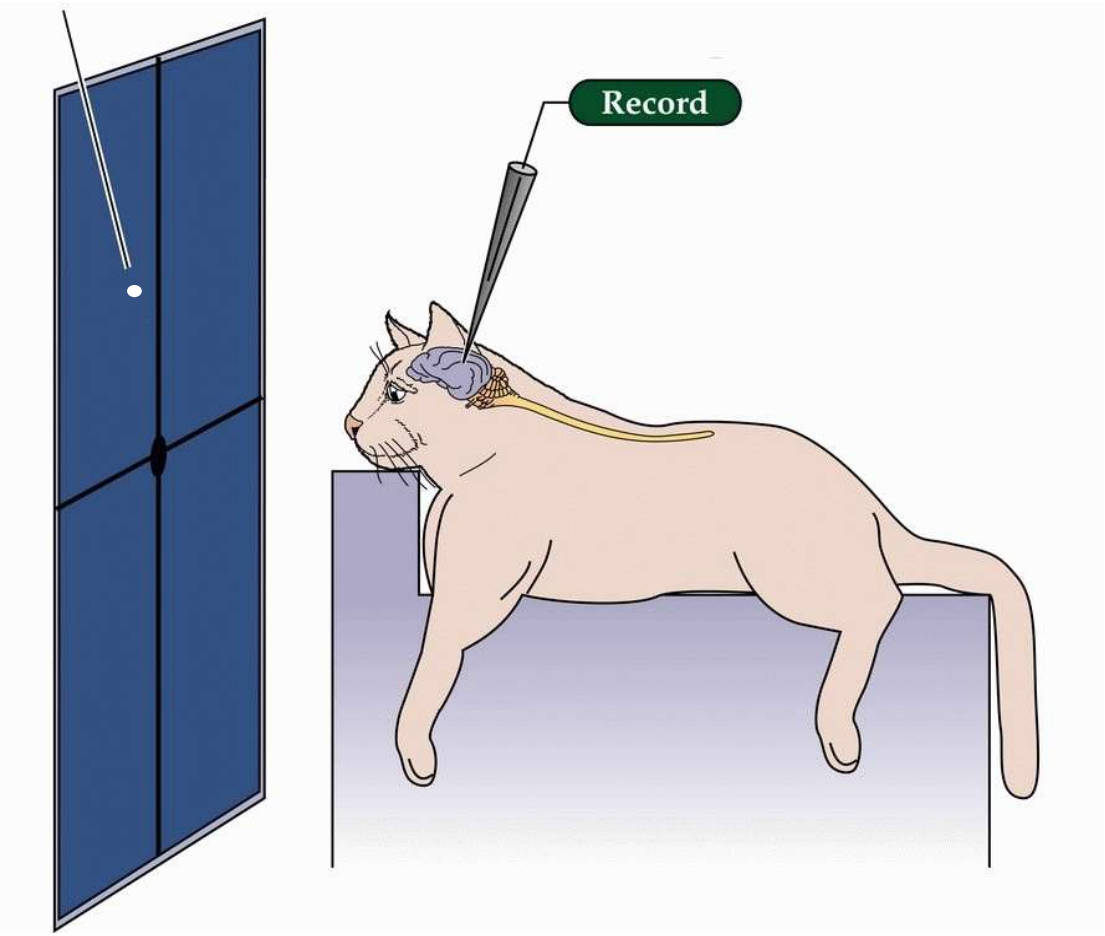


Primary visual pathway

The functional organization of the primary visual cortex

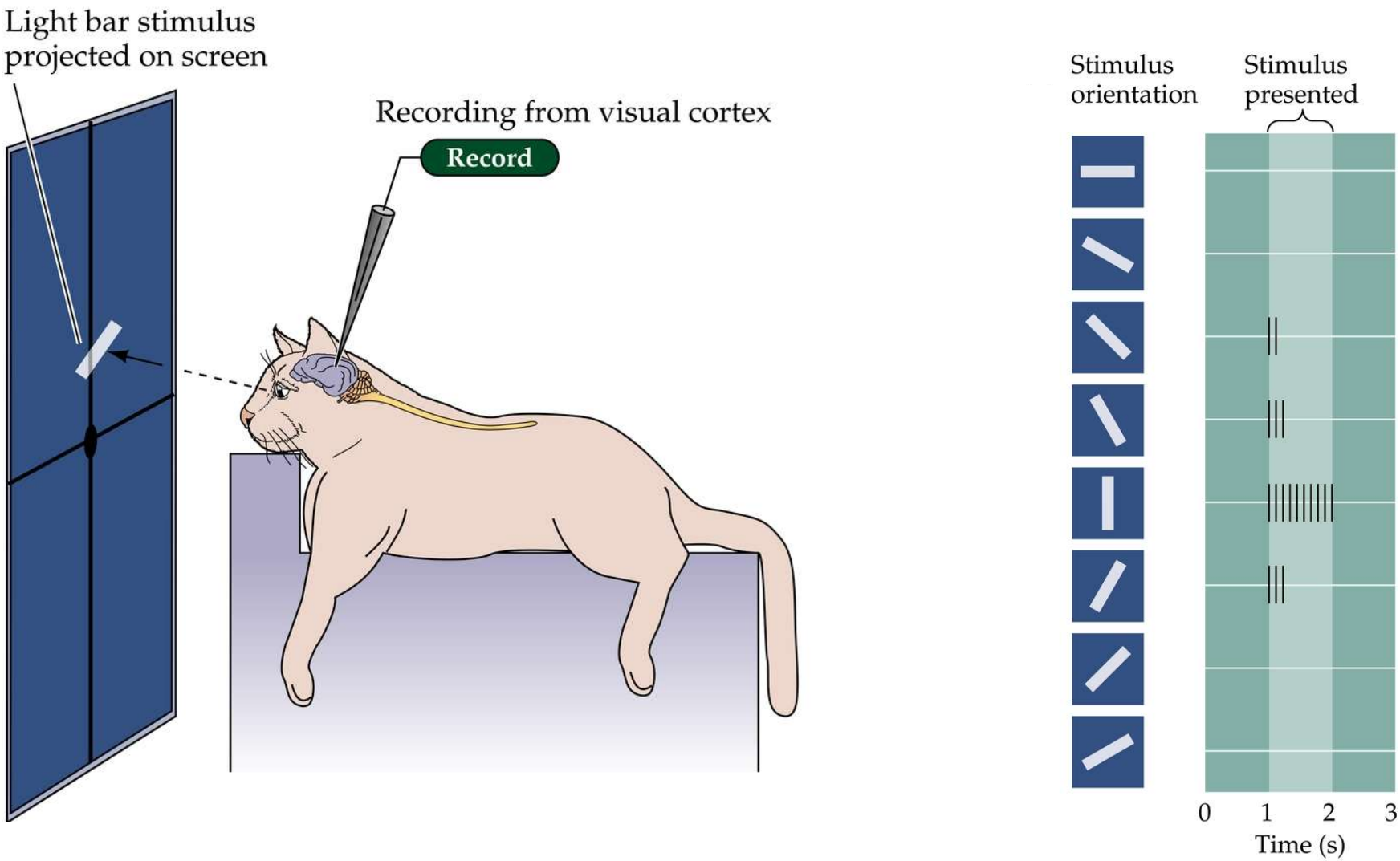
Point stimulation



David Hubel and **Torsten Wiesel**

Primary visual pathway

The functional organization of the primary visual cortex



Primary visual pathway

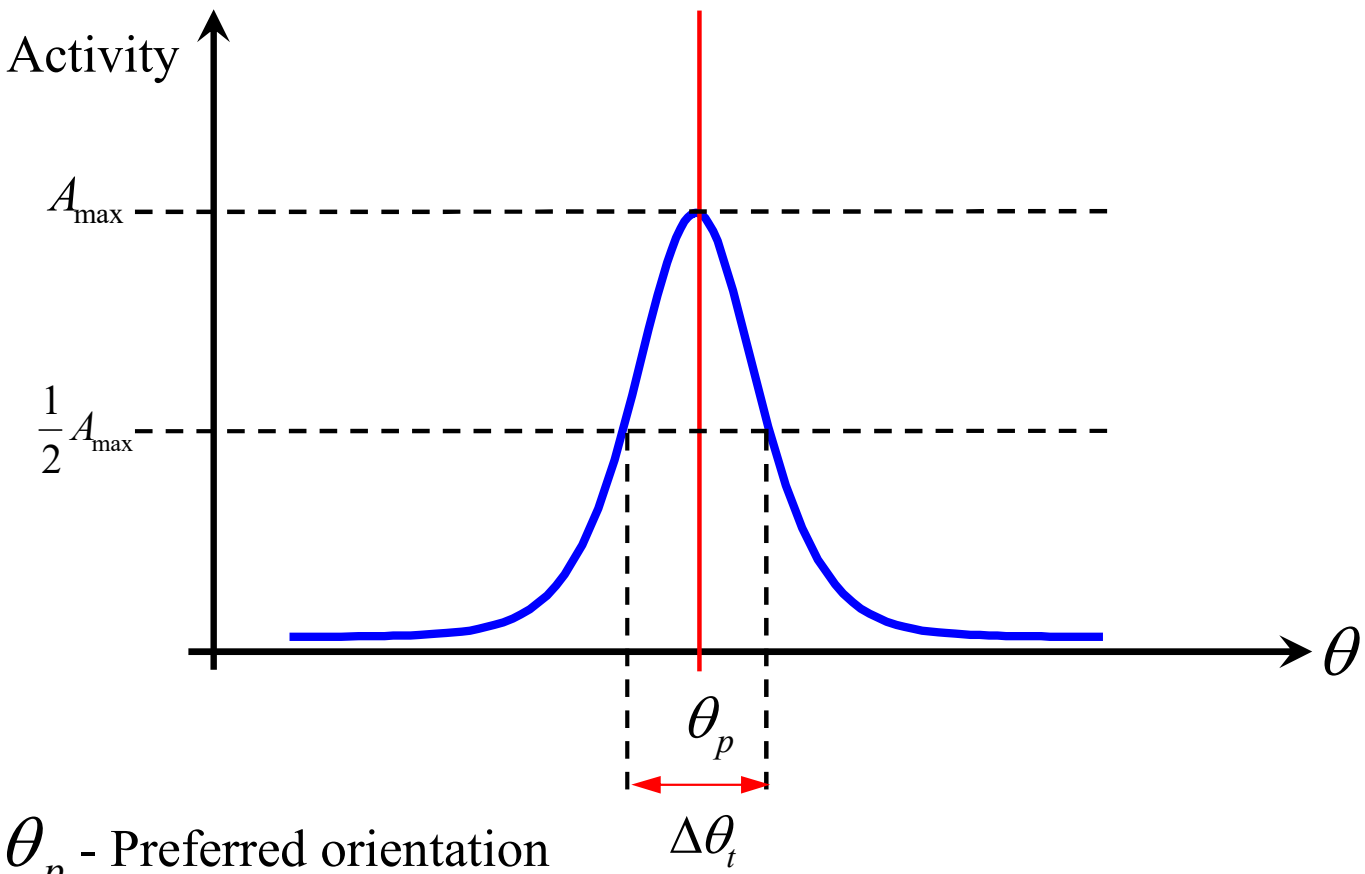
The functional organization of the primary visual cortex

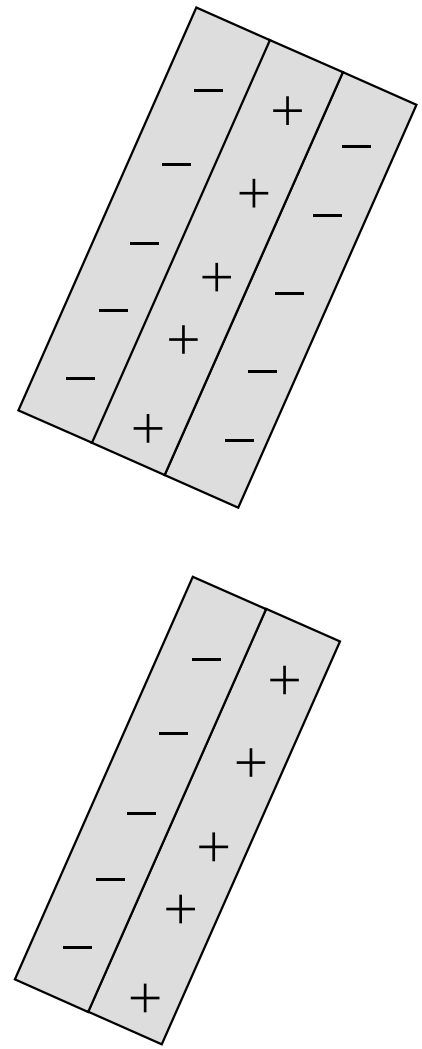
Hubel and Wiesel main findings:

1. Most cells in V1 have orientation preference.
2. V1 has three main types of cortical cells (according to their tuning properties) which were names **simple**, **complex**, and **hypercomplex** cells.
3. V1 cells exhibit all "shades" of ocular dominance.
4. Many V1 cells are also direction tuned (prefer moving stimuli in particular direction).
5. Most basic features are independent of visual experience (genetically determined).

Primary visual pathway

Orientation and spatial tuning of simple cells



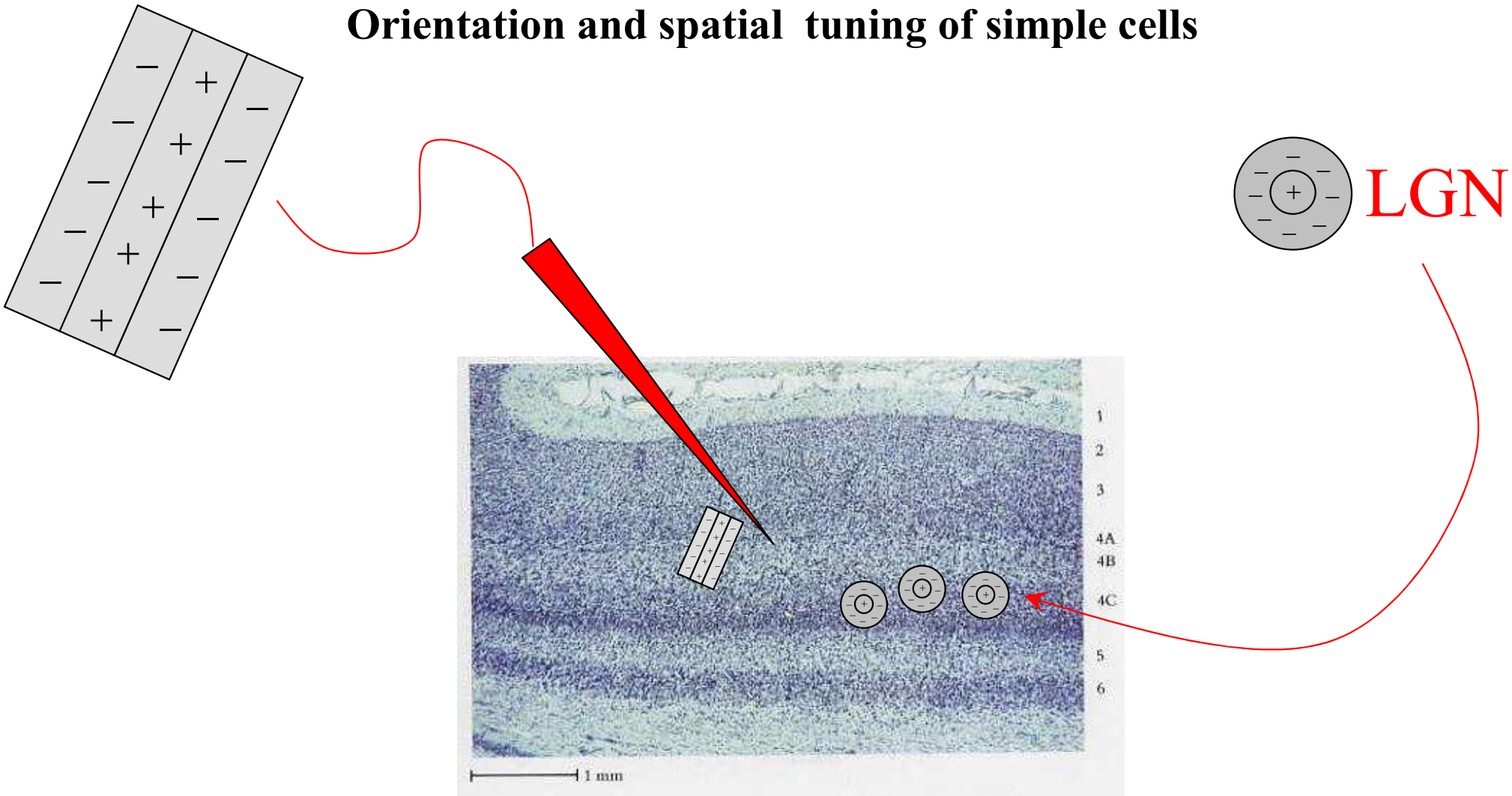


θ_p - Preferred orientation

$\Delta\theta_t$ - Orientation tuning

Primary visual pathway

Orientation and spatial tuning of simple cells



Primary visual pathway

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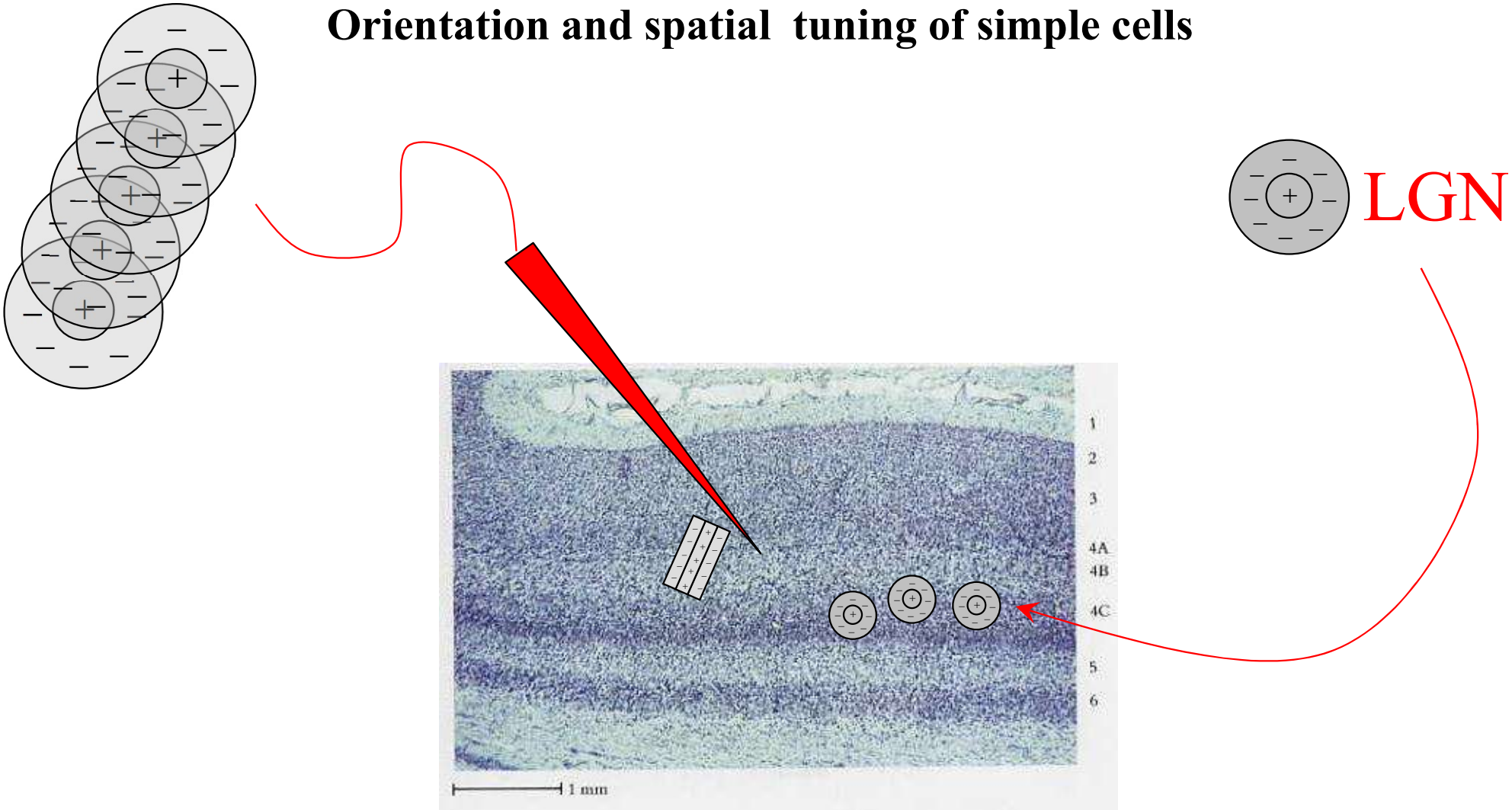
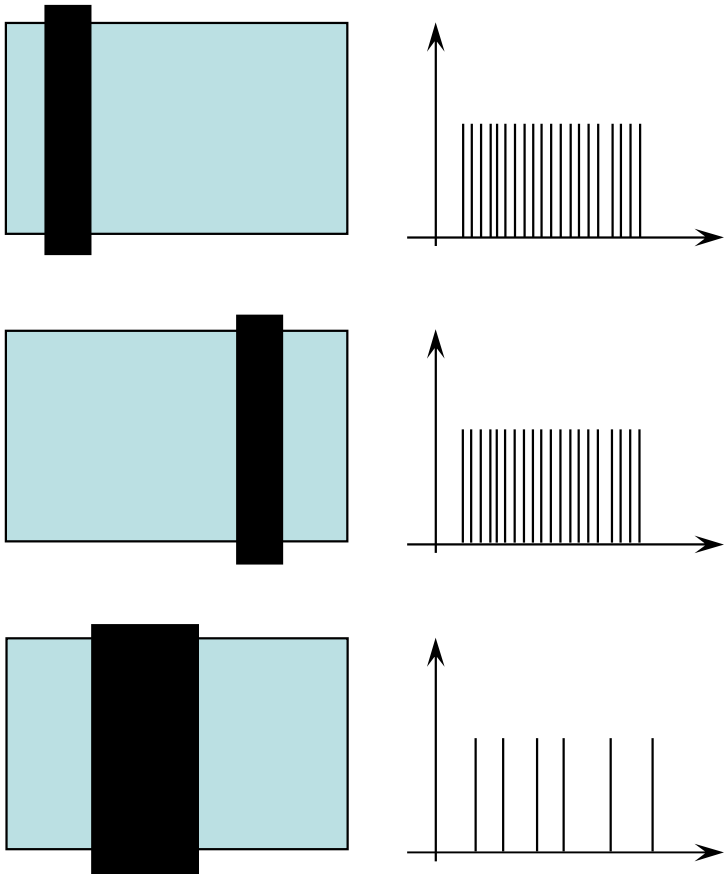


Primary visual pathway

Tuning properties of complex cells

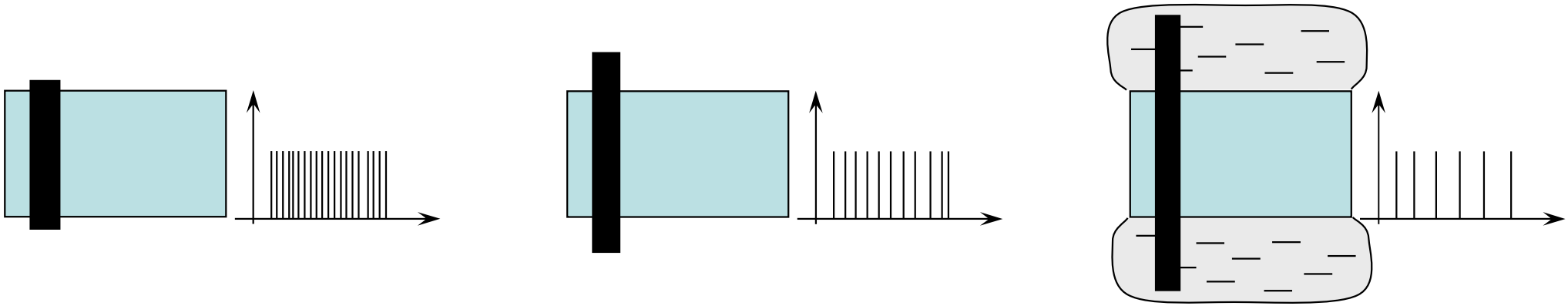
1. More than simple cells in numbers
2. Orientation selective
3. Large(r) spatial extension
4. Position insensitivity (within RF)
5. Motion sensitivity
6. Nonlinear response.



Primary visual pathway

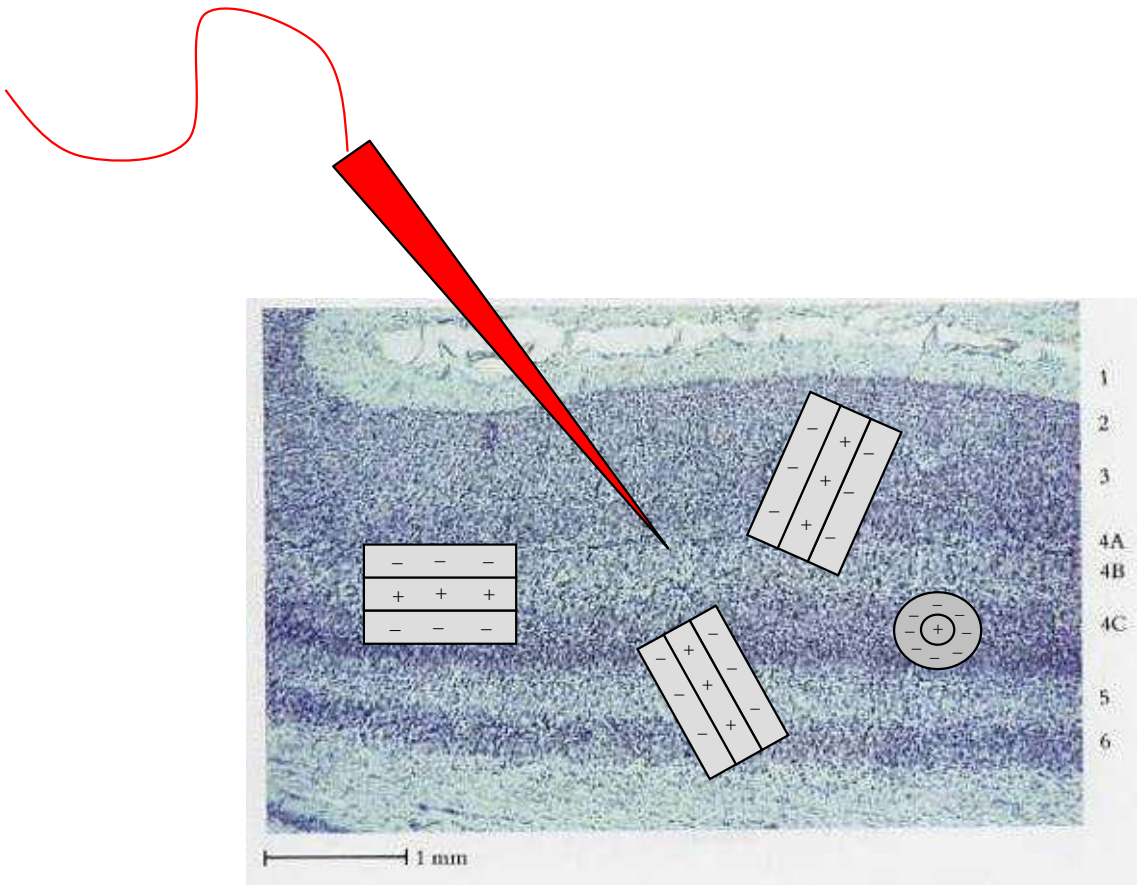
Tuning properties of hypercomplex cells

1. Resembles simple or complex cells in selectivity
2. Wide variety of subtypes depending on layers
   - variety of spatial extent
   - Sensitivity to wavelengths
3. End-stopped



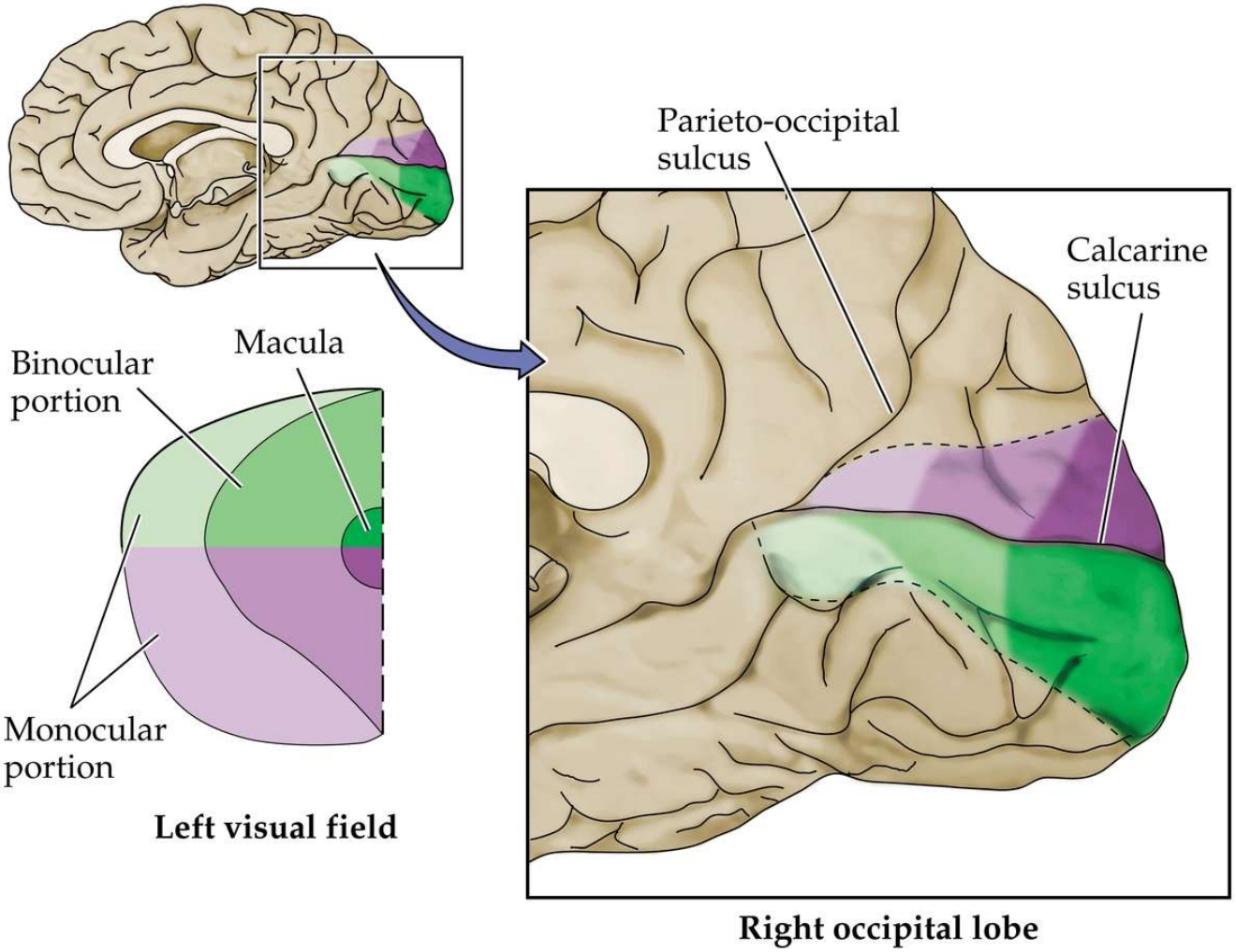
Primary visual pathway

Functional organization of the primary visual cortex



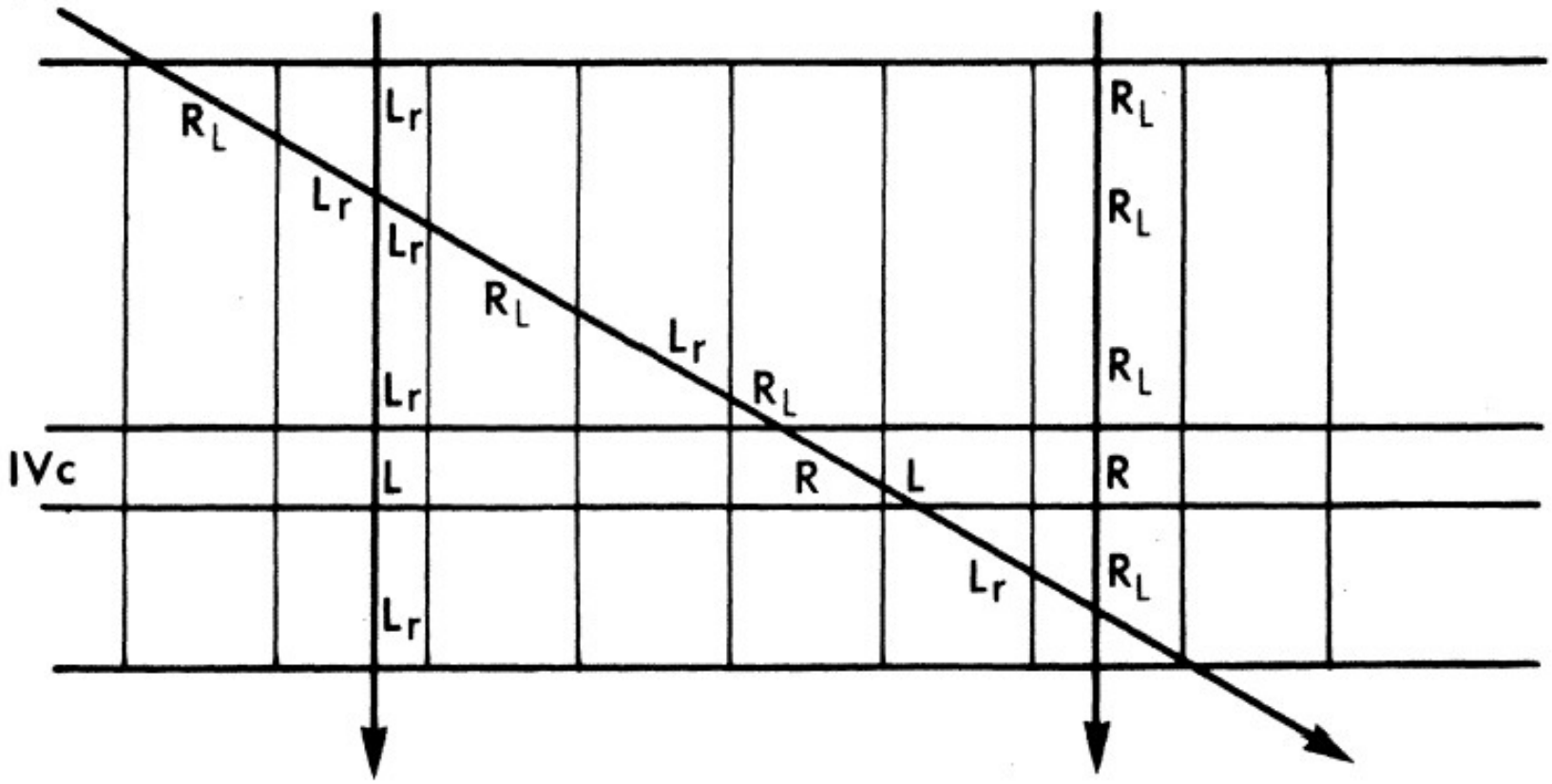
Primary visual pathway

The retinotopic map



Primary visual pathway

Ocular dominance slabs



Primary visual pathway

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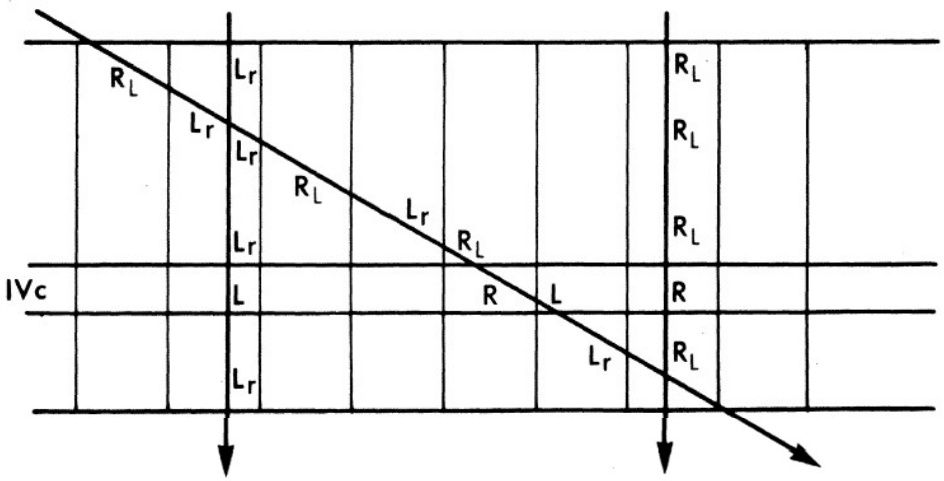

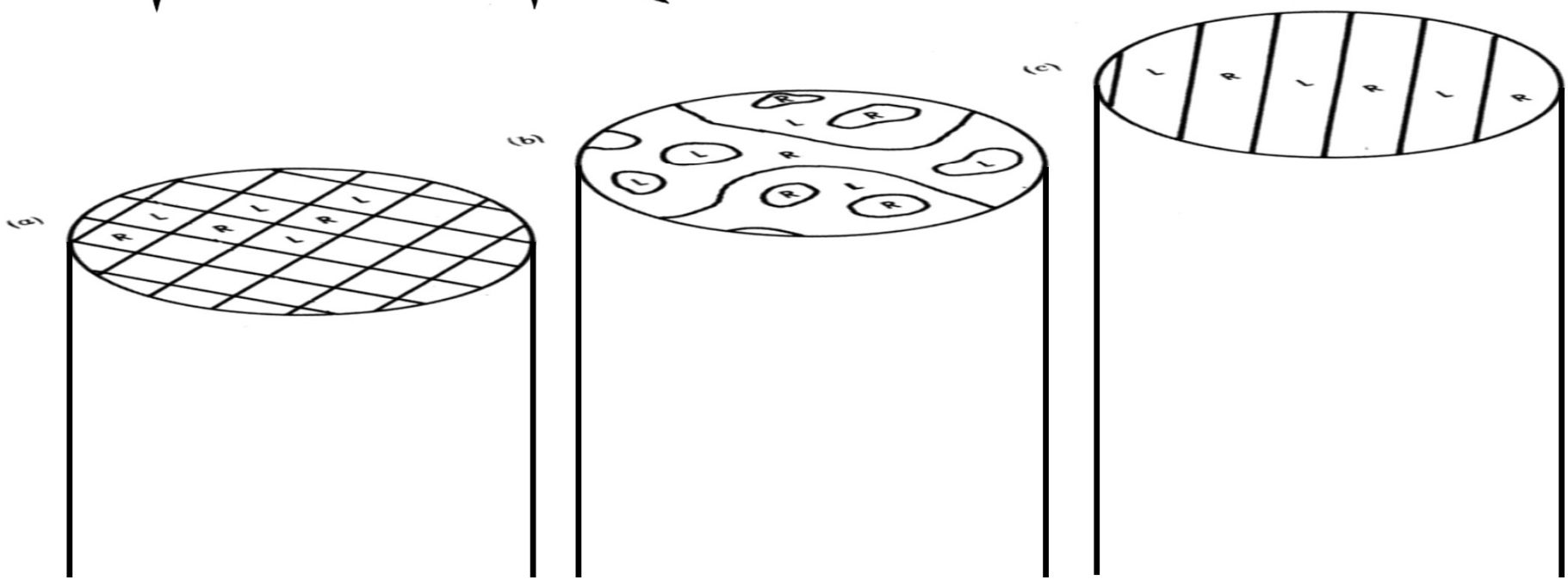

Primary visual pathway

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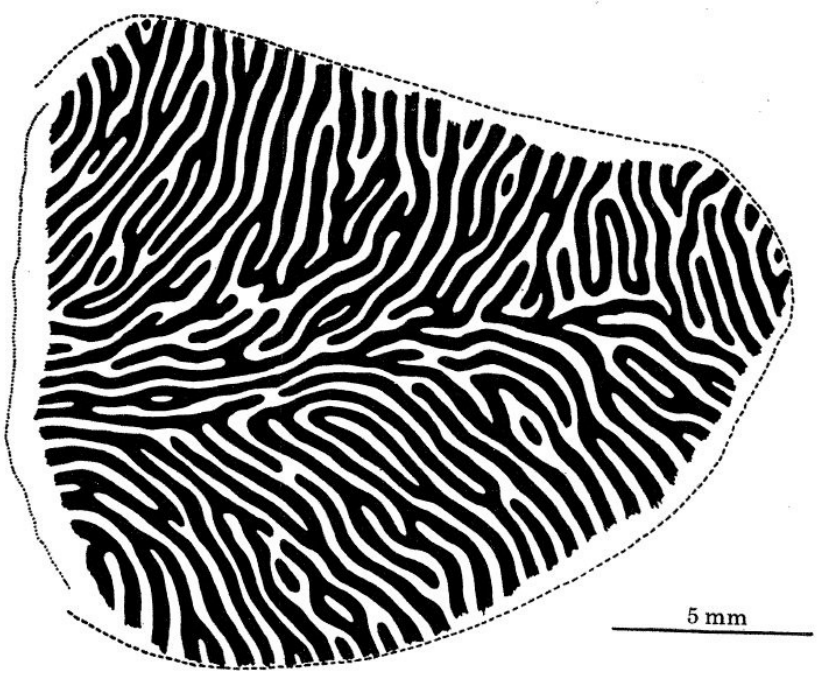
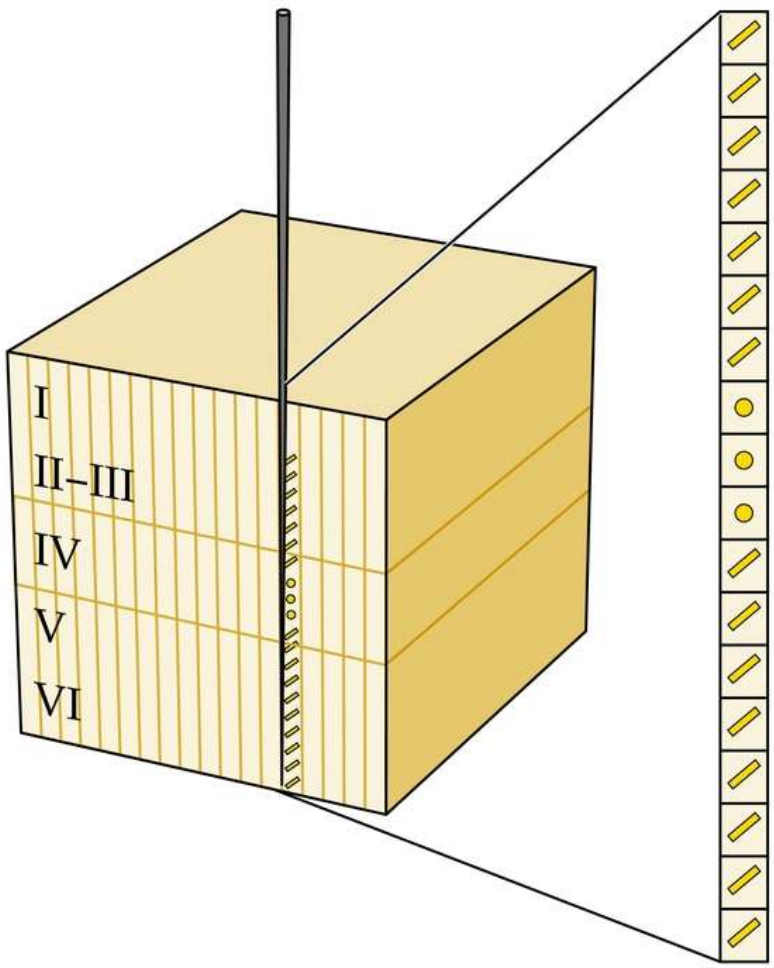


Primary visual pathway

Orientation columns



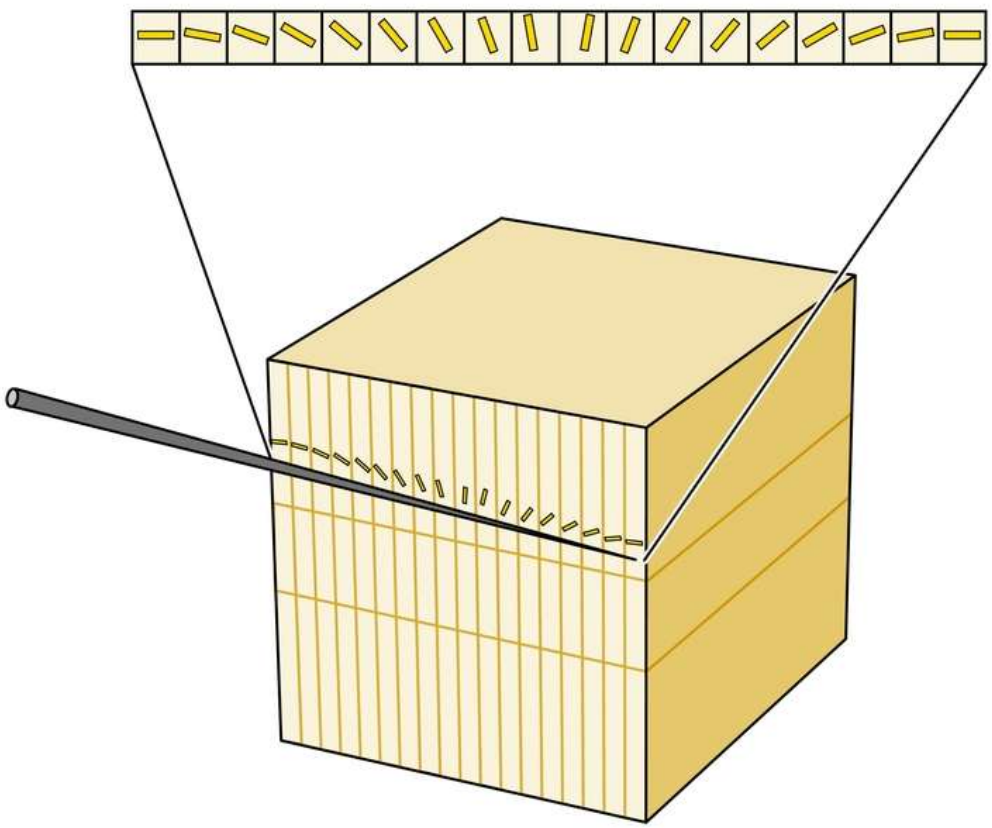
Electrode penetration perpendicular to the cortical surface encounters neurons of similar orientation preference

↓

V1 is a collection of orientation columns

Primary visual pathway

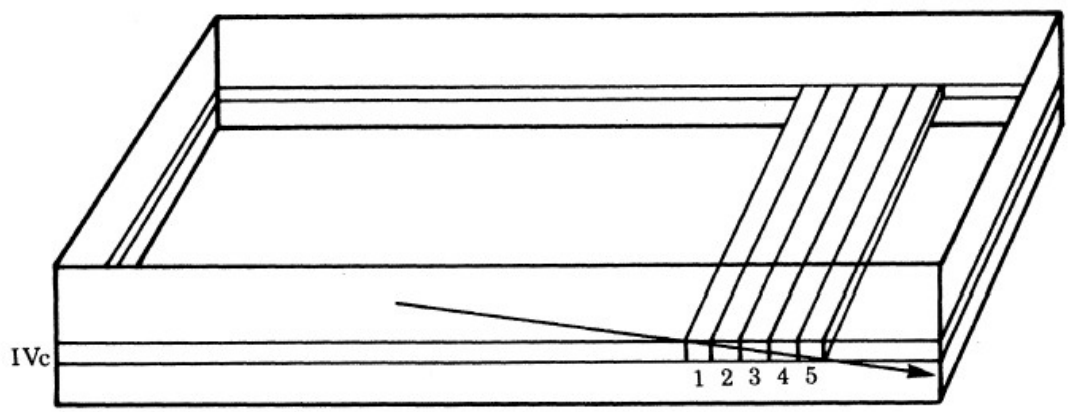
Orientation columns

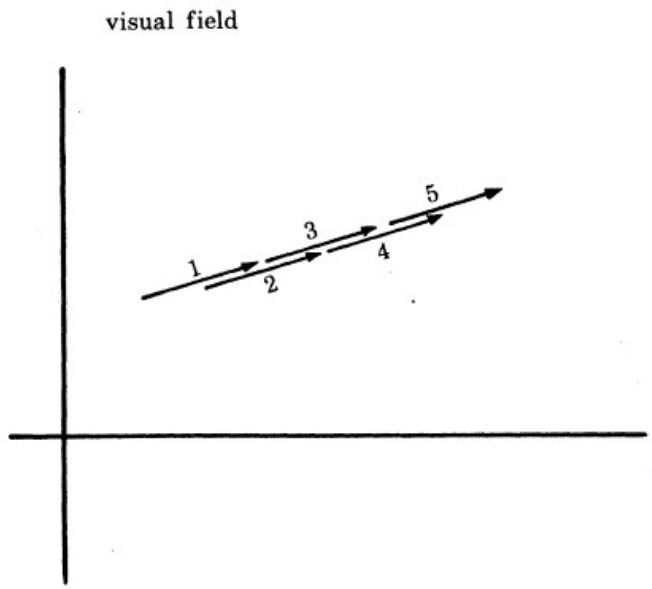


Electrode penetration parallel to the cortical surface encounters neurons whose orientation preference changes continuously

Primary visual pathway

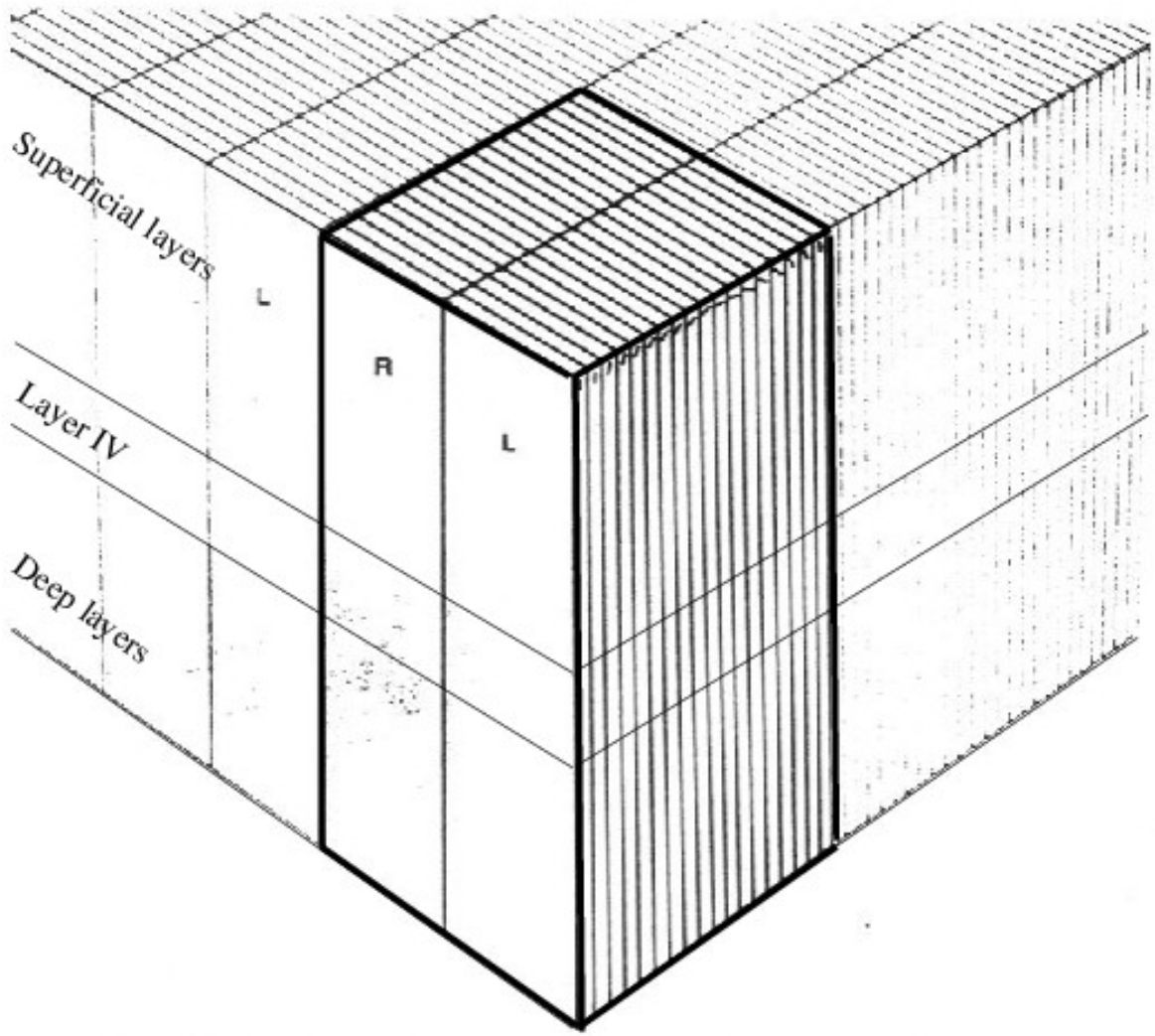
Spatial tuning progression





Primary visual pathway

Orientation hypercolumns



Orientation columns:

- ~50 μm
- ~10^0 resolution

Orientation hypercolumns:

- ~1 mm in width

Ocular dominance slabs:

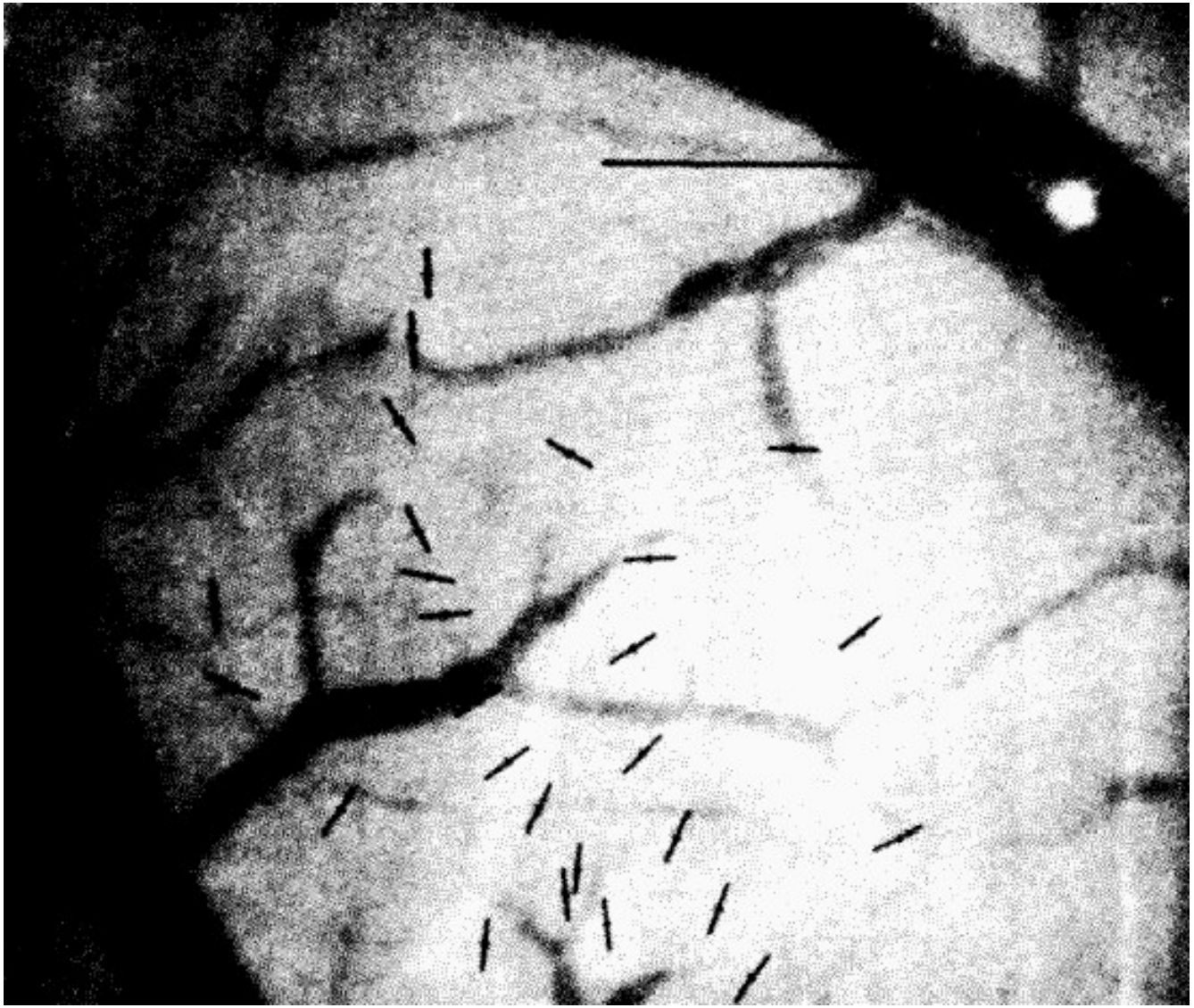
- ~0.5 mm in width

Basic cortical block:

- ~1mm x 1mm

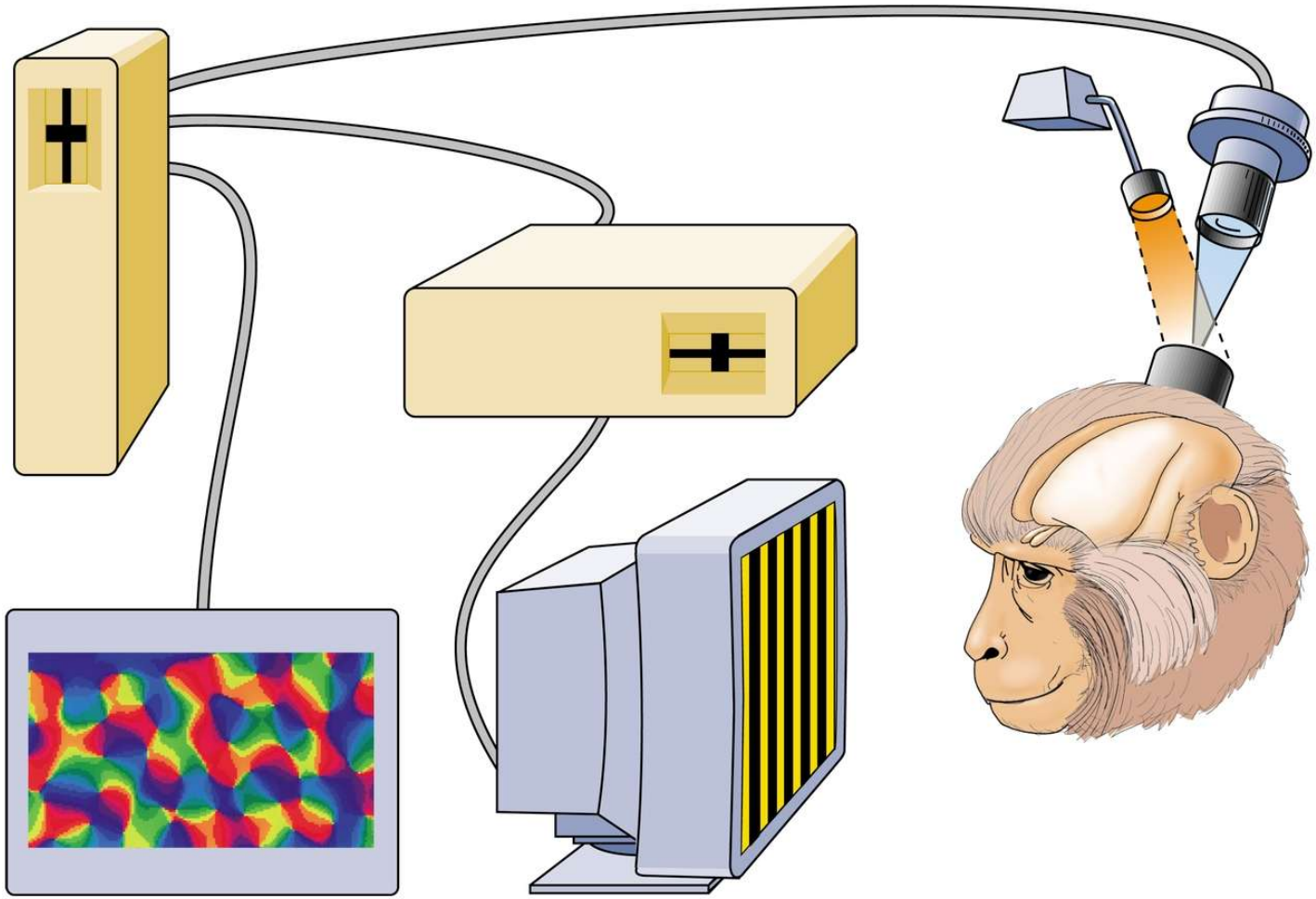
Primary visual pathway

Orientation hypercolumns



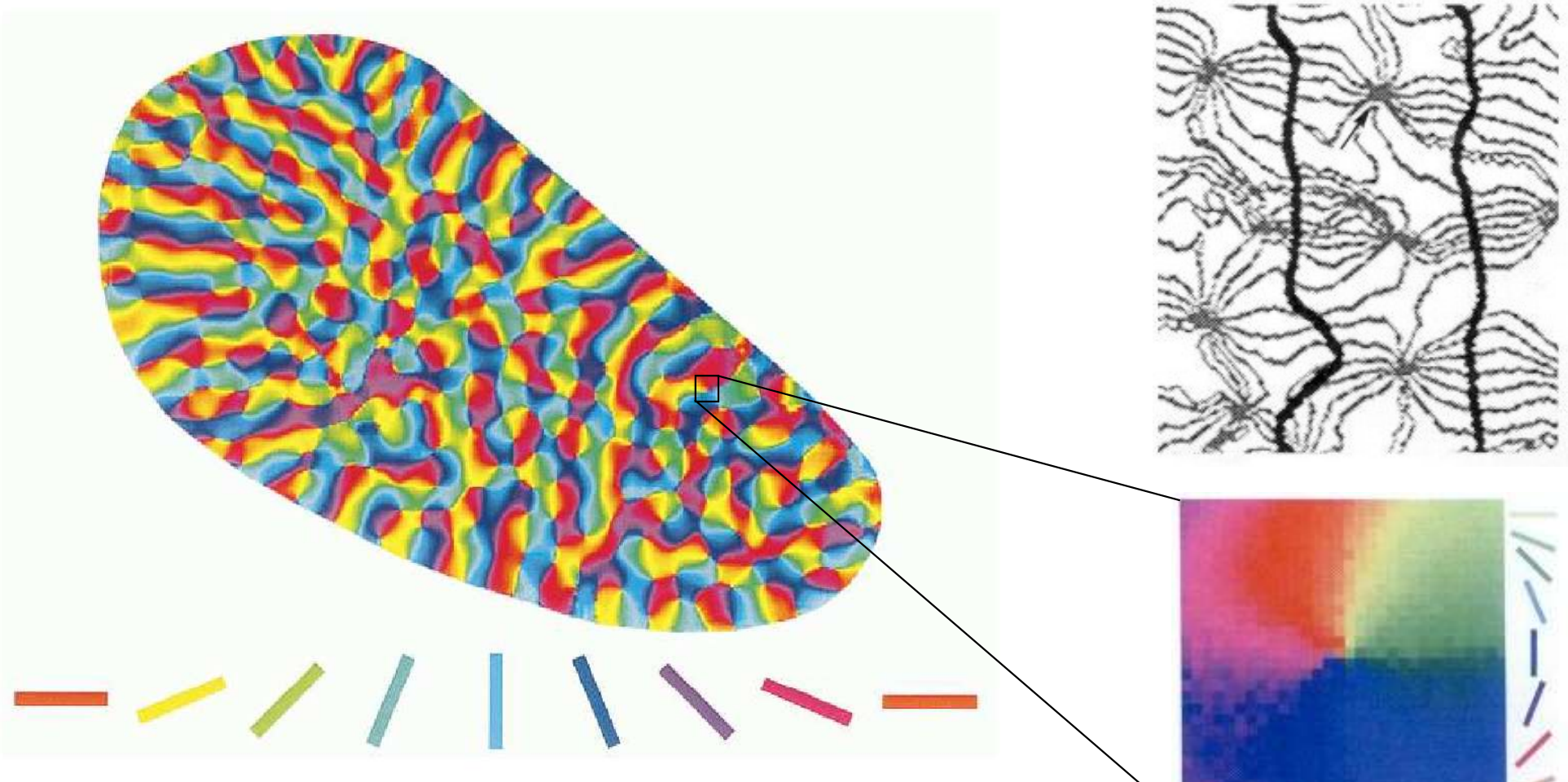
Primary visual pathway

Orientation hypercolumns – Optical imaging



Primary visual pathway

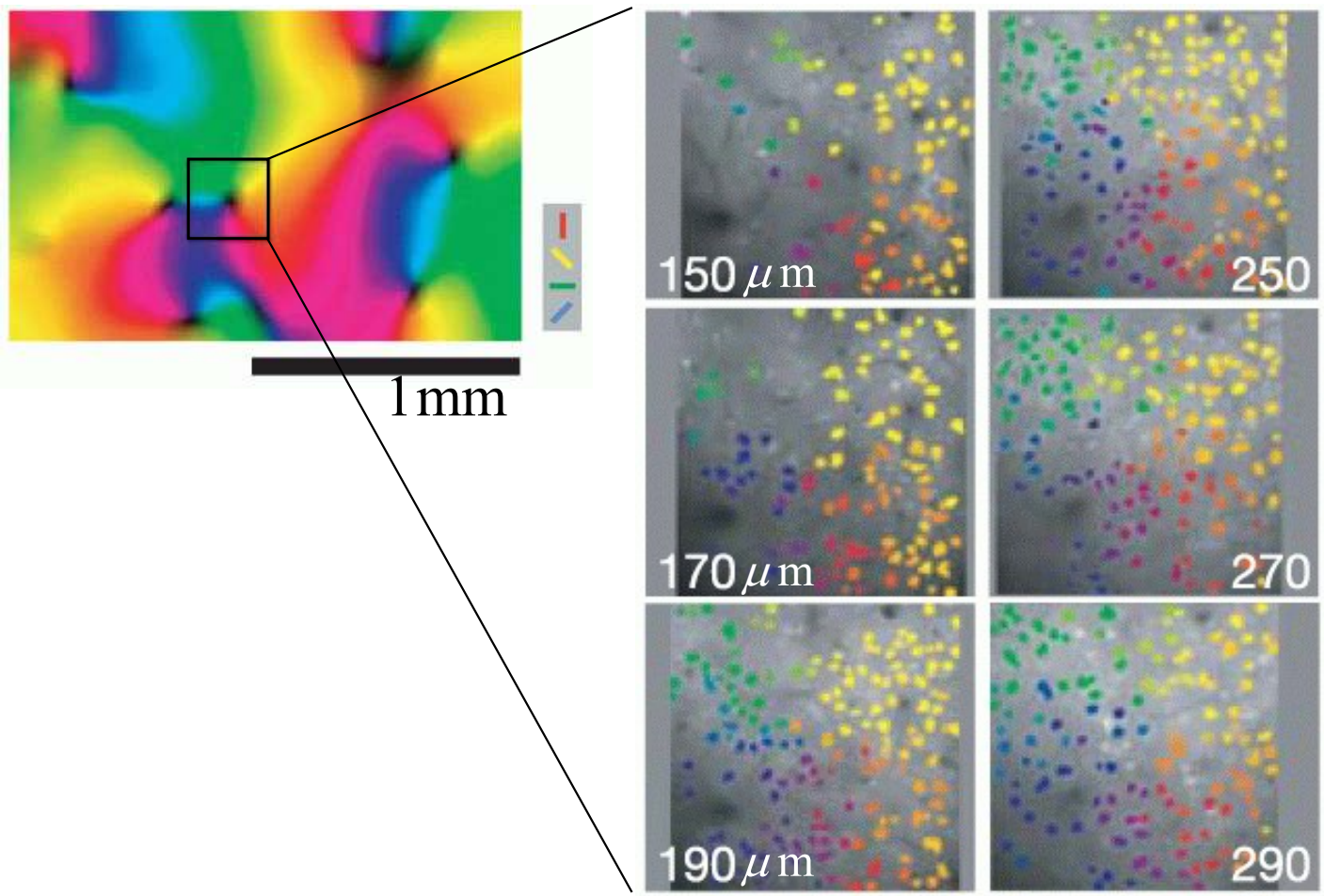
Orientation hypercolumns – Optical imaging

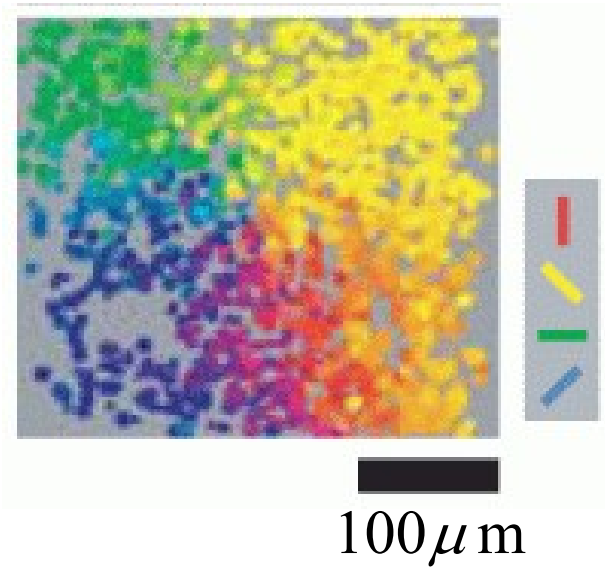


[Bosking *et al.*, 1997; after Grinvald *et al.*, 1986]

Primary visual pathway

Orientation hypercolumns – Two Photon Calcium Imaging

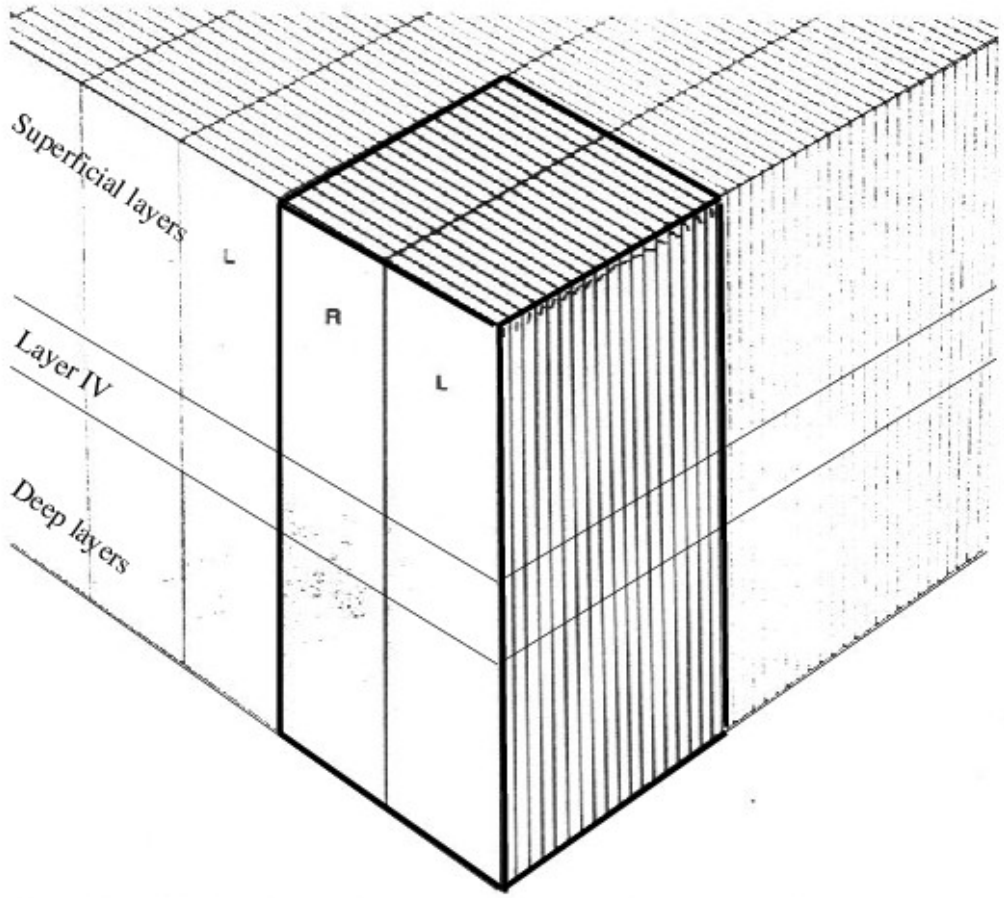




Ohki et al, *Nature* 2006.

Primary visual pathway

Computing in orientation columns



What can be computed within the ice cube model?

How?

RECORDING FROM
SINGLE NEURONS